

74. ON THE SIMULTANEOUS TRANSFORMATION OF TWO HOMOGENEOUS FUNCTIONS, &c. [pp. 429–431]

provided that we have

$$\begin{aligned} \mathfrak{a}_1 &= a\alpha^2 + b\alpha'^2 + c\alpha''^2 + 2f\alpha'\alpha'' + 2g\alpha''\alpha + 2h\alpha\alpha', \\ \mathfrak{b}_1 &= a\beta^2 + b\beta'^2 + c\beta''^2 + 2f\beta'\beta'' + 2g\beta''\beta + 2h\beta\beta', \\ \mathfrak{c}_1 &= a\gamma^2 + b\gamma'^2 + c\gamma''^2 + 2f\gamma'\gamma'' + 2g\gamma''\gamma + 2h\gamma\gamma', \\ \mathfrak{f}_1 &= a\beta\gamma + b\beta'\gamma' + c\beta''\gamma'' + f(\beta'\gamma'' + \beta''\gamma') + g(\beta''\gamma + \beta\gamma'') + h(\beta\gamma' + \beta'\gamma), \\ \mathfrak{g}_1 &= a\gamma\alpha + b\gamma'\alpha' + c\gamma''\alpha'' + f(\gamma'\alpha'' + \gamma''\alpha') + g(\gamma''\alpha + \gamma\alpha'') + h(\gamma\alpha' + \gamma'\alpha), \\ \mathfrak{h}_1 &= a\alpha\beta + b\alpha'\beta' + c\alpha''\beta'' + f(\alpha'\beta'' + \alpha''\beta') + g(\alpha''\beta + \alpha\beta'') + h(\alpha\beta' + \alpha'\beta). \end{aligned}$$

Representing for a moment the equations between the pairs of functions of the second order by

$$u = u_1, \quad U = U_1, \quad v = v_1,$$

we have, whatever be the value of λ ,

$$\lambda u + U + v = \lambda u_1 + U_1 + v_1.$$

Hence, if

$$\begin{vmatrix} \alpha, & \beta, & \gamma \\ \alpha', & \beta', & \gamma' \\ \alpha'', & \beta'', & \gamma'' \end{vmatrix} = \Omega; \quad \text{then}$$

$$\begin{vmatrix} \lambda\mathfrak{a}_1 + A_1 + \mathfrak{a}_1, & \lambda\mathfrak{h}_1 + H_1 + \mathfrak{h}_1, & \lambda\mathfrak{g}_1 + G_1 + \mathfrak{g}_1 \\ \lambda\mathfrak{h}_1 + H_1 + \mathfrak{h}_1, & \lambda\mathfrak{b}_1 + B_1 + \mathfrak{b}_1, & \lambda\mathfrak{f}_1 + F_1 + \mathfrak{f}_1 \\ \lambda\mathfrak{g}_1 + G_1 + \mathfrak{g}_1, & \lambda\mathfrak{f}_1 + F_1 + \mathfrak{f}_1, & \lambda\mathfrak{c}_1 + C_1 + \mathfrak{c}_1 \end{vmatrix} = \Omega^2 \begin{vmatrix} \lambda a + A + \mathfrak{a}, & \lambda h + H + \mathfrak{h}, & \lambda g + G + \mathfrak{g} \\ \lambda h + H + \mathfrak{h}, & \lambda b + B + \mathfrak{b}, & \lambda f + F + \mathfrak{f} \\ \lambda g + G + \mathfrak{g}, & \lambda f + F + \mathfrak{f}, & \lambda c + C + \mathfrak{c} \end{vmatrix}$$

Hence, since a, b, c, f, g, h are arbitrary,

$$\begin{vmatrix} \lambda\mathfrak{a}_1 + A_1, & \lambda\mathfrak{h}_1 + H_1, & \lambda\mathfrak{g}_1 + G_1 \\ \lambda\mathfrak{h}_1 + H_1, & \lambda\mathfrak{b}_1 + B_1, & \lambda\mathfrak{f}_1 + F_1 \\ \lambda\mathfrak{g}_1 + G_1, & \lambda\mathfrak{f}_1 + F_1, & \lambda\mathfrak{c}_1 + C_1 \end{vmatrix} = \Omega^2 \begin{vmatrix} \lambda a + A, & \lambda h + H, & \lambda g + G \\ \lambda h + H, & \lambda b + B, & \lambda f + F \\ \lambda g + G, & \lambda f + F, & \lambda c + C \end{vmatrix}$$

which determine the relations between the coefficients of the functions of the second order. When we derive

$$\begin{vmatrix} \mathfrak{a}_1, & \mathfrak{h}_1, & \mathfrak{g}_1 \\ \mathfrak{h}_1, & \mathfrak{b}_1, & \mathfrak{f}_1 \\ \mathfrak{g}_1, & \mathfrak{f}_1, & \mathfrak{c}_1 \end{vmatrix} = \Omega^2 \begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix}$$

and by comparing the coefficients of λ , &c., if we write for shortness,

$$\mathfrak{a} = \begin{vmatrix} \lambda b + B, & \lambda f + F \\ \lambda f + F, & \lambda c + C \end{vmatrix}, \quad \&c., \quad \text{then}$$

[continuation of large equation block, p. 430]

$$\begin{aligned} \mathfrak{a}_1\alpha^2 + \mathfrak{b}_1\beta^2 + \mathfrak{c}_1\gamma^2 + 2\mathfrak{f}_1\beta\gamma + 2\mathfrak{g}_1\gamma\alpha + 2\mathfrak{h}_1\alpha\beta &= \Omega^2\mathfrak{a}, \\ \mathfrak{a}_1\alpha'^2 + \mathfrak{b}_1\beta'^2 + \mathfrak{c}_1\gamma'^2 + 2\mathfrak{f}_1\beta'\gamma' + 2\mathfrak{g}_1\gamma'\alpha' + 2\mathfrak{h}_1\alpha'\beta' &= \Omega^2\mathfrak{b}, \\ \mathfrak{a}_1\alpha''^2 + \mathfrak{b}_1\beta''^2 + \mathfrak{c}_1\gamma''^2 + 2\mathfrak{f}_1\beta''\gamma'' + 2\mathfrak{g}_1\gamma''\alpha'' + 2\mathfrak{h}_1\alpha''\beta'' &= \Omega^2\mathfrak{c}, \\ \mathfrak{a}_1\alpha'\alpha'' + \mathfrak{b}_1\beta'\beta'' + \mathfrak{c}_1\gamma'\gamma'' + \mathfrak{f}_1(\beta'\gamma'' + \beta''\gamma') + \mathfrak{g}_1(\gamma'\alpha'' + \gamma''\alpha') + \mathfrak{h}_1(\alpha'\beta'' + \alpha''\beta') &= \Omega^2\mathfrak{f}, \\ \mathfrak{a}_1\alpha''\alpha + \mathfrak{b}_1\beta''\beta + \mathfrak{c}_1\gamma''\gamma + \mathfrak{f}_1(\beta''\gamma + \beta\gamma'') + \mathfrak{g}_1(\gamma''\alpha + \gamma\alpha'') + \mathfrak{h}_1(\alpha''\beta + \alpha\beta'') &= \Omega^2\mathfrak{g}, \\ \mathfrak{a}_1\alpha\alpha' + \mathfrak{b}_1\beta\beta' + \mathfrak{c}_1\gamma\gamma' + \mathfrak{f}_1(\beta\gamma' + \beta'\gamma) + \mathfrak{g}_1(\gamma\alpha' + \gamma'\alpha) + \mathfrak{h}_1(\alpha\beta' + \alpha'\beta) &= \Omega^2\mathfrak{h}, \end{aligned}$$

each of which virtually contains three equations on account of the indeterminate quantity λ . A somewhat more elegant form may be given to these equations; thus the first of them is

$$\begin{vmatrix} \alpha, & \beta, & \gamma \\ \alpha, & \lambda \mathbf{a}_1 + A_1, & \lambda \mathbf{h}_1 + H_1, & \lambda \mathbf{g}_1 + G_1 \\ \beta, & \lambda \mathbf{h}_1 + H_1, & \lambda \mathbf{b}_1 + B_1, & \lambda \mathbf{f}_1 + F_1 \\ \gamma, & \lambda \mathbf{g}_1 + G_1, & \lambda \mathbf{f}_1 + F_1, & \lambda \mathbf{c}_1 + C_1 \end{vmatrix} = \Omega^2 \begin{vmatrix} \cdot, & \cdot, & \cdot \\ \cdot, & \lambda b + B, & \lambda f + F \\ \cdot, & \lambda f + F, & \lambda c + C \end{vmatrix},$$

from which the form of the whole system is sufficiently obvious. The actual values of the coefficients α, β , &c. can only be obtained in the particular case where

$$\mathbf{f}_1 = \mathbf{g}_1 = \mathbf{h}_1 = F_1 = G_1 = H_1 = 0.$$

If we suppose besides (which is no additional loss of generality) that $\mathbf{a}_1 = \mathbf{b}_1 = \mathbf{c}_1 = 1$, then the whole system of formulae becomes

$$(A_1 + \lambda)(B_1 + \lambda)(C_1 + \lambda) = \Omega^2 \begin{vmatrix} \lambda a + A, & \lambda h + H, & \lambda g + G \\ \lambda h + H, & \lambda b + B, & \lambda f + F \\ \lambda g + G, & \lambda f + F, & \lambda c + C \end{vmatrix}$$

or $\Omega^2 = \kappa^{-1}$ suppose; and then

$$\begin{aligned} (B_1 + \lambda)(C_1 + \lambda)\alpha^2 + (C_1 + \lambda)(A_1 + \lambda)\beta^2 + (A_1 + \lambda)(B_1 + \lambda)\gamma^2 &= \frac{1}{\kappa} \mathbf{a}, \\ (B_1 + \lambda)(C_1 + \lambda)\alpha'^2 + (C_1 + \lambda)(A_1 + \lambda)\beta'^2 + (A_1 + \lambda)(B_1 + \lambda)\gamma'^2 &= \frac{1}{\kappa} \mathbf{b}, \\ (B_1 + \lambda)(C_1 + \lambda)\alpha''^2 + (C_1 + \lambda)(A_1 + \lambda)\beta''^2 + (A_1 + \lambda)(B_1 + \lambda)\gamma''^2 &= \frac{1}{\kappa} \mathbf{c}, \\ (B_1 + \lambda)(C_1 + \lambda)\alpha'\alpha'' + (C_1 + \lambda)(A_1 + \lambda)\beta'\beta'' + (A_1 + \lambda)(B_1 + \lambda)\gamma'\gamma'' &= \frac{1}{\kappa} \mathbf{f}, \\ (B_1 + \lambda)(C_1 + \lambda)\alpha''\alpha + (C_1 + \lambda)(A_1 + \lambda)\beta''\beta + (A_1 + \lambda)(B_1 + \lambda)\gamma''\gamma &= \frac{1}{\kappa} \mathbf{g}, \\ (B_1 + \lambda)(C_1 + \lambda)\alpha\alpha' + (C_1 + \lambda)(A_1 + \lambda)\beta\beta' + (A_1 + \lambda)(B_1 + \lambda)\gamma\gamma' &= \frac{1}{\kappa} \mathbf{h}; \end{aligned}$$

p. 431. where, writing down the expanded values of $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{f}, \mathbf{g}, \mathbf{h}$:

$$\begin{aligned} (\lambda b + B)(\lambda c + C) - (\lambda f + F)^2 &= \mathbf{a}, \\ (\lambda c + C)(\lambda a + A) - (\lambda g + G)^2 &= \mathbf{b}, \\ (\lambda a + A)(\lambda b + B) - (\lambda h + H)^2 &= \mathbf{c}; \\ (\lambda g + G)(\lambda h + H) - (\lambda a + A)(\lambda f + F) &= \mathbf{f}, \\ (\lambda h + H)(\lambda f + F) - (\lambda b + B)(\lambda g + G) &= \mathbf{g}, \\ (\lambda f + F)(\lambda g + G) - (\lambda c + C)(\lambda h + H) &= \mathbf{h}. \end{aligned}$$

By writing successively $\lambda = -A_1$, $\lambda = -B_1$, $\lambda = -C_1$, we see in the first place that A_1, B_1, C_1 are the roots of the same cubic equation, and we obtain next the values of $\alpha^2, \beta^2, \gamma^2$, &c. in terms of these quantities A_1, B_1, C_1 , and of the coefficients a, b , &c., A, B , &c. It is easy to see how the above formulae would have been modified if $\mathbf{a}_1, \mathbf{b}_1, \mathbf{c}_1$, instead of being equal to unity, had one or more of them been equal to unity with a negative sign. It is obvious that every step of the preceding process is equally applicable whatever be the number of variables.

75. ON THE ATTRACTION OF AN ELLIPSOID.

[From the *Cambridge and Dublin Mathematical Journal*, vol. iv. (1849), pp. 50–65.]

PART I.—ON LEGENDRE'S SOLUTION OF THE PROBLEM OF THE ATTRACTION OF AN ELLIPSOID ON AN EXTERNAL POINT.

I propose in the following paper to give an outline of Legendre's investigation of the attraction of an ellipsoid upon an exterior point, ["Mémoire sur les Intégrales Doubles," Paris, Mém. Acad. Sc. for 1788, published 1791, pp. 454–486], one of the earliest and (notwithstanding its complexity) most elegant

solutions of the problem. It will be convenient to begin by considering some of the geometrical properties of a system of cones made use of in the investigation.

§ 1. The equation of the ellipsoid referred to axes parallel to the principal axes, and passing through the attracted point, may be written under the form

$$l(x-a)^2 + m(y-b)^2 + n(z-c)^2 - k = 0,$$

(where $\sqrt{k/l}$, $\sqrt{k/m}$, $\sqrt{k/n}$ are the semiaxes, and a, b, c are the coordinates of the attracted point referred to the principal axes). Or putting $la^2 + mb^2 + nc^2 - k = \delta$, this equation becomes

$$lx^2 + my^2 + nz^2 - 2(lax + mby + ncz) + \delta = 0.$$

The cones in question are those which have the same axes and directions of circular section as the cone having its vertex in the attracted point and circumscribed about the ellipsoid. The equation of the system of cones (containing the arbitrary parameter ω) is

$$(lx^2 + my^2 + nz^2)\delta - (lax + mby + ncz)^2 + \omega^2(x^2 + y^2 + z^2) = 0;$$

or as it may also be written,

$$(\omega^2 + l\delta - l^2a^2)x^2 + (\omega^2 + m\delta - m^2b^2)y^2 + (\omega^2 + n\delta - n^2c^2)z^2 - 2mnbcyz - 2nlcazx - 2lmabxy = 0.$$

p. 433. For $\omega = 0$, the cone coincides with the circumscribed cone; as ω increases, the aperture of the cone gradually diminishes, until for a certain value, $\omega = \Omega$, the cone reduces itself to a straight line (the normal of the confocal ellipsoid through the attracted point). It is easily seen that Ω is the positive root of the equation

$$\frac{l^2a^2}{\Omega^2 + l\delta} + \frac{m^2b^2}{\Omega^2 + m\delta} + \frac{n^2c^2}{\Omega^2 + n\delta} = 1,$$

a different form of which may be obtained by writing $\Omega^2 = k\theta/\xi$, ξ being then determined by means of the equation

$$\frac{la^2}{k + l\xi} + \frac{mb^2}{k + m\xi} + \frac{nc^2}{k + n\xi} = 1;$$

that is,

$$\sqrt{(k/l + \xi)}, \quad \sqrt{(k/m + \xi)}, \quad \sqrt{(k/n + \xi)}$$

are the semiaxes of the confocal ellipsoid through the attracted point.

In the case where ω remains indeterminate, it is obvious that the cone intersects the ellipsoid in the curve in which the ellipsoid is intersected by a certain hyperboloid of revolution of two sheets, having the attracted point for a focus, and the plane of contact of the ellipsoid with the circumscribed cone (that is the polar plane of the attracted point) for the corresponding directrix plane: also the excentricity of the hyperboloid is $\frac{1}{\omega}\sqrt{(l^2a^2 + m^2b^2 + n^2c^2)}$, which suffices for its complete determination. For $\omega = 0$, the hyperboloid reduces itself to the plane of contact of the ellipsoid with the circumscribed cone, and for $\omega = \Omega$, the hyperboloid and the ellipsoid have a double contact, viz. at the points where the ellipsoid is intersected by the normal to the confocal ellipsoid through the attracted point.

If ω remains constant while k is supposed to vary, that is, if the ellipsoid vary in magnitude (the position and proportion of its axes remaining unaltered), the locus of the intersection of the cone and the ellipsoid is a surface of the fourth order defined by the equation

$$(lx^2 + my^2 + nz^2 - lax - mby - ncz)^2 = \omega^2(x^2 + y^2 + z^2),$$

and consisting of an exterior and an interior sheet meeting at the attracted point, which is a conical point on the surface, viz. a point where the tangent plane is replaced by a tangent cone. The general form of this surface is easily seen from the figure, in which the ellipsoid has been replaced by a sphere, and the surface in question

[Figure: a sphere centred near the attracted point M with the fourth-order surface generated by revolution shown in cross-section, axis CM .]

p. 434. is that generated by the revolution of the curve round the line CM . The surface of the fourth order being once described for any particular value of ω , the cone corresponding to any one of the series of similar, similarly situated, and concentric ellipsoids is at once determined by means of the intersection of the ellipsoid in question with the surface of the fourth order. It is clear too that there is always one of these ellipsoids which has a double contact with the surface of the fourth order, viz. at the points where this ellipsoid is intersected by the normal to the confocal ellipsoid through the attracted point; thus there is always an ellipsoid for which the cone corresponding to a given value of ω reduces itself to a straight line.

Consider the attracting ellipsoid, which for distinction may be termed the ellipsoid S , and the two cones C, C' , which correspond to the values $\omega, \omega - d\omega$ of the variable parameter. Legendre shows that the attraction of the portion of the ellipsoid S included between the two cones C, C' is independent of the quantity k , which determines the magnitude of the ellipsoid: that is, if there be any other ellipsoid T similarly situated and concentric to and with the ellipsoid S , and two cones D, D' , which for the ellipsoid T correspond to the same values $\omega, \omega - d\omega$ of the variable parameter; then the attraction of the portion of the ellipsoid S , included between the two cones C, C' , is equal to the attraction of the portion of the ellipsoid T included between the two cones D and D' . By taking for the ellipsoid T the ellipsoid for which the cone D reduces itself to a straight line, the aperture of the cone D' is indefinitely small, and the attraction of the portion of the ellipsoid T included within the cone D' is at once determined; and thus the attraction of the portion of the ellipsoid S included between the cones C, C' is obtained in a finite form. Hence the attraction of the portion of the ellipsoid S included between any two cones C_l, C_n , corresponding to the values ω_l, ω_n , of the variable parameter, is expressed by means of a single integral, and by extending the integration from $\omega = 0$ to $\omega = \sqrt{(k\theta/\xi)}$, the attraction of the whole ellipsoid is obtained in the form of a single integral readily reducible to that given by the ordinary solutions. It is clear too that the attraction of the portion of the ellipsoid S included between any two cones C_l, C_n , is equal to that of the portion of the ellipsoid T included between the corresponding cones D_l and D_n . Hence also, assuming for the ellipsoid T , that for which the cone D_n reduces itself to a straight line, and supposing that the cones C_l and D_l coincide with the circumscribing cones, the attraction of the portion of the ellipsoid S exterior to the cone C_n is equal to the attraction of the entire ellipsoid T . More generally, the attraction of the portion of the ellipsoid S included between the cones C_l and C_n is equal to the attraction of the shell included between the surfaces of the two ellipsoids, for which the cones D_l and D_n respectively reduce themselves to straight lines.

§ 2. Proceeding to the analytical solution, and resuming the equation of the ellipsoid

$$lx^2 + my^2 + nz^2 - 2(lax + mby + ncz) + \delta = 0,$$

and that of the cone

$$(lx^2 + my^2 + nz^2)\delta - (lax + mby + ncz)^2 + \omega^2(x^2 + y^2 + z^2) = 0;$$

p. 435. consider a radius vector on the conical surface such that the cosines of its inclinations to the axes are

$$\frac{P}{\Theta}, \quad \frac{Q}{\Theta}, \quad \frac{R}{\Theta}, \quad (\Theta = \sqrt{(P^2 + Q^2 + R^2)})$$

P, Q, R and Θ being functions of the parameter ω , and of another variable ϕ , which determines the position of the radius vector upon the conical surface. Also let ρ be the length of the portion of the radius vector which lies within the ellipsoid; then representing by dS the spherical angle corresponding to the variations of ω and ϕ , the attraction in the direction of the axis of x is given by the formula

$$A = \iint \rho \frac{P}{\Theta} dS.$$

Also by a known formula

$$dS = \frac{1}{\Theta^3} \left[P \left(\frac{dQ}{d\phi} \frac{dR}{d\omega} - \frac{dR}{d\phi} \frac{dQ}{d\omega} \right) + Q \left(\frac{dR}{d\phi} \frac{dP}{d\omega} - \frac{dP}{d\phi} \frac{dR}{d\omega} \right) + R \left(\frac{dP}{d\phi} \frac{dQ}{d\omega} - \frac{dQ}{d\phi} \frac{dP}{d\omega} \right) \right] d\phi d\omega,$$

and it is easy to obtain

$$\rho = \frac{2\omega^2}{lP^2 + mQ^2 + nR^2}.$$

The quantities P, Q, R have now to be expressed as functions of ω, ϕ , so that their values substituted for x, y, z , may satisfy identically the equation of the cone. This may be done by assuming

$$\begin{aligned} P &= \omega p, \\ Q &= mb(\omega^2 + n\delta)\left(la + \frac{p}{U} \cos \phi\right) - \frac{D}{U} \sqrt{nc} \omega \sin \phi, \\ R &= nc(\omega^2 + m\delta)\left(la + \frac{p}{U} \cos \phi\right) + \frac{D}{U} \sqrt{mb} \omega \sin \phi, \end{aligned}$$

where

$$\begin{aligned} p &= (\omega^2 + m\delta)(\omega^2 + n\delta) - m^2b^2(\omega^2 + n\delta) - n^2c^2(\omega^2 + m\delta), \\ U^2 &= m^2b^2(\omega^2 + n\delta) + n^2c^2(\omega^2 + m\delta), \\ D^2 &= (\omega^2 + l\delta)(\omega^2 + m\delta)(\omega^2 + n\delta) \left[\frac{l^2a^2}{\omega^2 + l\delta} + \frac{m^2b^2}{\omega^2 + m\delta} + \frac{n^2c^2}{\omega^2 + n\delta} - 1 \right]; \end{aligned}$$

a system of values which, in point of fact, depend upon the following geometrical considerations: by treating x as a constant in the equation of the cone, that is, in effect by considering the sections of the cone by planes parallel to that of yz , the equation of the cone becomes that of an ellipse; transforming first to a set of axes through the centre and then to a set of conjugate axes, one of which passes through the point where the plane of the ellipse is intersected by the axis of x , then the equation takes the form $\frac{\eta^2}{A^2} + \frac{\zeta^2}{B^2} = 1$, and is satisfied by $\eta = A \cos \phi$, $\zeta = B \sin \phi$, A and B being of course linear functions of these values, the preceding expressions may be obtained.

p. 436. The substitution of the above values of P, Q, R (a somewhat tedious one which does not occur in the process actually made use of by Legendre) gives the very simple result,

$$dS = \frac{P^2 \omega d\omega d\phi}{\Theta^3};$$

and the formula for the attraction becomes

$$A = 2 \iint \frac{P^3 \omega^2 d\omega d\phi}{lP^2 + mQ^2 + nR^2},$$

which is of the form $A = 2 \int I \omega^2 d\omega$, where

$$I = \int \frac{P^3 d\phi}{lP^2 + mQ^2 + nR^2},$$

which last integral, taken between the limits $\phi = 0$ and $\phi = 2\pi$, and multiplied by $2\omega^2 d\omega$, expresses the attraction of the portion of the ellipsoid included between two consecutive cones. The integration is evidently possible, but the actual performance of it is the great difficulty of Legendre's process. The result, as before mentioned, is independent of the quantity k , or, what comes to the same thing, of the quantity δ : assuming this property (an assumption which in fact resolves itself into the consideration of the ellipsoid for which the cone reduces itself to a straight line, as before explained), the integral is at once obtained by writing $\delta = \Delta$ where Δ represents the positive root of the equation

$$\frac{l^2a^2}{\omega^2 + l\Delta} + \frac{m^2b^2}{\omega^2 + m\Delta} + \frac{n^2c^2}{\omega^2 + n\Delta} - 1 = 0.$$

This gives

$$P = \frac{l^2a^2}{\omega^2}(\omega^2 + m\Delta)(\omega^2 + n\Delta), \quad Q = lmab(\omega^2 + n\Delta), \quad R = lnac(\omega^2 + m\Delta),$$

values independent of ϕ , or the value of I is found by multiplying the quantity under the integral sign by 2π ; and hence we have

$$A = 4\pi la \int \frac{l^2a^2(\omega^2 + l\Delta)^2(\omega^2 + m\Delta)^2(\omega^2 + n\Delta)^2 d\omega}{l^3a^2(\omega^2 + m\Delta)^2(\omega^2 + n\Delta)^2 + m^3b^2(\omega^2 + n\Delta)^2(\omega^2 + l\Delta)^2 + n^3c^2(\omega^2 + l\Delta)^2(\omega^2 + m\Delta)^2},$$

where of course Δ is to be considered as a function of ω . By integrating from $\omega = 0$, to $\omega = \omega_r$, we have the attraction of the portion of the ellipsoid included between any two of the series of cones, and to obtain

the attraction of the whole ellipsoid we must integrate from $\omega = 0$ to $\omega = \sqrt{(k\theta/\xi)}$, where ξ is determined as before by the equation

$$\frac{la^2}{k+l\xi} + \frac{mb^2}{k+m\xi} + \frac{nc^2}{k+n\xi} = 1;$$

p. 437. and it is obvious that for this value of ω we have $\Delta = \delta$. The expression for the attraction is easily reduced to a known form by writing $\omega^2 = k\theta/\xi$; this gives

$$A = 4\pi la \int \frac{k^3(k+ly)^2(k+my)^2(k+ny)^2 a d\omega}{l^3 a^2(k+my)^2(k+ny)^2 + m^3 b^2(k+ny)^2(k+ly)^2 + n^3 c^2(k+ly)^2(k+my)^2}.$$

Also

$$\omega^2 = k \left(\frac{l^2 a^2}{k+ly} + \frac{m^2 b^2}{k+my} + \frac{n^2 c^2}{k+ny} \right),$$

whence

$$\omega d\omega = - \frac{k [l n a^2 (k+my)^2 (k+ny)^2 + m^3 b^2 (k+ly)^2 (k+ny)^2 + n^3 c^2 (k+ly)^2 (k+my)^2] dy}{2(k+ly)^2 (k+my)^2 (k+ny)^2},$$

and thus

$$A = 2\pi k^2 la \int \frac{dy}{(k+ly)^{3/2} (k+my)^{1/2} (k+ny)^{1/2}},$$

where for the entire ellipsoid the integral is to be taken from $y = \xi$ to $y = \infty$. A better known form is readily obtained by writing $x^2 = \frac{k+l\xi}{k+ly}$, in which case the limits for the entire ellipsoid are $x = 0$, $x = 1$.

It may be as well to indicate the first step of the reduction of the integral I , viz. the method of resolving the denominator into two factors. We have identically,

$$(\Delta - \delta)(lP^2 + mQ^2 + nR^2) = \omega^2(P^2 + Q^2 + R^2) + \Delta(lP^2 + mQ^2 + nR^2) - (laP + mbQ + ncR)^2,$$

and the second side of this equation is resolvable into two factors independently of the particular values of P, Q, R . Representing this second side for a moment in the notation of a general quadratic function, or under the form

$$AP^2 + BQ^2 + CR^2 + 2FQR + 2GRP + 2HPQ,$$

we have the required solution,

$$lP^2 + mQ^2 + nR^2 = \frac{1}{\mathcal{A}} [\mathcal{A}P + \{\mathcal{H} + \sqrt{(-\mathfrak{C})}\}Q + \{\mathcal{G} + \sqrt{(-\mathfrak{B})}\}R] [\mathcal{A}P + \{\mathcal{H} - \sqrt{(-\mathfrak{C})}\}Q + \{\mathcal{G} - \sqrt{(-\mathfrak{B})}\}R];$$

where, as usual, $\mathfrak{B} = \mathcal{C}\mathcal{A} - \mathcal{G}^2$, $\mathfrak{C} = \mathcal{A}\mathcal{B} - \mathcal{H}^2$, and the roots must be so taken that $\sqrt{(-\mathfrak{B})}\sqrt{(-\mathfrak{C})} = \mathfrak{F} \{\mathfrak{F} = (\mathcal{G}\mathcal{H} - \mathcal{A}\mathcal{F})\}$.

I have purposely restricted myself so far to the problem considered by Legendre: the general transformation, of which the preceding is a particular case, and also a simpler mode of effecting the integration, are given in the next part of this paper.

p. 438.

PART II.—ON A FORMULA FOR THE TRANSFORMATION OF CERTAIN MULTIPLE INTEGRALS.

Consider the integral

$$V = \int F(x, y, \dots) dx dy \dots,$$

where the number of variables $x, y, \dots$ is equal to n , and $F(x, y, \dots)$ is a homogeneous function of the order μ .

Suppose that $x, y, \dots$ are connected by a homogeneous equation $\psi(x, y, \dots) = 0$ containing a variable parameter ω (so that ω is a homogeneous function of the order zero in the variables $x, y, \dots$). Then, writing

$$r^2 = x^2 + y^2 + \dots, \quad x = r\alpha, \quad y = r\beta, \dots,$$

the quantities $\alpha, \beta, \dots$ are connected by the equations

$$\alpha^2 + \beta^2 + \dots = 1, \quad \psi(\alpha, \beta, \dots) = 0,$$

and we may therefore consider them as functions of ω and of $(n-2)$ independent variables θ, ϕ , &c.; whence

$$dx dy \dots = r^{n-1} \nabla dr d\omega d\theta \dots,$$

where

$$\nabla = \begin{vmatrix} \alpha, & \beta, & \dots \\ \frac{d\alpha}{d\omega}, & \frac{d\beta}{d\omega}, & \\ \frac{d\alpha}{d\theta}, & \frac{d\beta}{d\theta}, & \\ \vdots & \vdots & \end{vmatrix}.$$

Also

$$F(x, y, \dots) = r^\mu F(\alpha, \beta, \dots),$$

and therefore

$$V = \int r^{\mu+n-1} F(\alpha, \beta, \dots) \nabla dr d\omega d\theta \dots,$$

or, integrating with respect to r ,

$$\int r^{\mu+n-1} dr = \frac{1}{\mu+n} r^{\mu+n},$$

which, taken between the proper limits, is a function of $\alpha, \beta, \dots$, equal $f(\alpha, \beta, \dots)$ suppose; this gives

$$V = \int f(\alpha, \beta, \dots) F(\alpha, \beta, \dots) \nabla d\omega d\theta \dots,$$

in which I shall assume that the limits of ω are constant. If, in order to get rid of the condition $\alpha^2 + \beta^2 + \dots = 1$, we assume

$$\alpha = \frac{p}{\rho}, \quad \beta = \frac{q}{\rho}, \dots, \quad \rho^2 = p^2 + q^2 + \dots,$$

p. 439. the preceding expression for V becomes

$$V = \int f\left(\frac{p}{\rho}, \frac{q}{\rho}, \dots\right) F(p, q, \dots) \frac{1}{\rho^{\mu+n}} D d\omega d\theta \dots,$$

in which

$$D = \begin{vmatrix} p, & q, & \dots \\ \frac{dp}{d\omega}, & \frac{dq}{d\omega}, & \\ \frac{dp}{d\theta}, & \frac{dq}{d\theta}, & \\ \vdots & \vdots & \end{vmatrix}.$$

Assume

$$\begin{aligned} p &= P\xi + P'\eta + P''\zeta + \dots, \\ q &= Q\xi + Q'\eta + Q''\zeta + \dots, \\ &\vdots \end{aligned}$$

where the number of variables $\xi, \eta, \zeta, \dots$ (functions in general of ω, θ , &c.) is n , and where the coefficients P, Q , &c. are supposed to be functions of ω only. We have

$$\frac{dp}{d\theta} = P \frac{d\xi}{d\theta} + P' \frac{d\eta}{d\theta} + P'' \frac{d\zeta}{d\theta} + \dots, \quad \frac{dq}{d\theta} = Q \frac{d\xi}{d\theta} + Q' \frac{d\eta}{d\theta} + Q'' \frac{d\zeta}{d\theta} + \dots,$$

and, substituting these values as well as those of p, q , &c., but retaining the terms $\frac{dp}{d\omega}, \frac{dq}{d\omega}$, &c. in their original form, the determinant D resolves itself into the sum of a series of products,

$$\begin{vmatrix} \frac{dp}{d\omega}, & \frac{dq}{d\omega}, & \dots \\ P', & Q', & \dots \\ P'', & Q'', & \dots \\ \vdots & \vdots & \end{vmatrix} \cdot \begin{vmatrix} 1, & \cdot, & \dots \\ \cdot, & \eta, & \zeta, \dots \\ \cdot, & \frac{d\eta}{d\theta}, & \frac{d\zeta}{d\theta}, \dots \\ \vdots & \vdots & \end{vmatrix}.$$

Let Ψ be the function to which $\psi(p, q, \dots)$ is changed by the substitution of the above values of $p, q, \dots$; so that Ψ is a homogeneous function of $\xi, \eta, \zeta, \dots$ and we have the relation $\Psi = 0$. (It will be convenient to consider $\xi, \eta, \zeta, \dots$ as functions of ω, θ , &c., such as to satisfy identically this last equation.) We deduce

$$\frac{1}{X} \begin{vmatrix} 1, & \dots \\ \eta, & \zeta, \dots \\ \frac{d\eta}{d\theta}, & \frac{d\zeta}{d\theta}, \dots \\ \vdots & \end{vmatrix} = \frac{1}{Y} \begin{vmatrix} \cdot, & 1, & \dots \\ \xi, & \cdot, & \zeta, \dots \\ \frac{d\xi}{d\theta}, & \cdot, & \frac{d\zeta}{d\theta}, \dots \\ \vdots & \end{vmatrix} = \dots = S \text{ suppose,}$$

p. 440. where for shortness $X = \frac{d\Psi}{d\xi}$, $Y = \frac{d\Psi}{d\eta}$, &c. The substitution of these values gives

$$D = S \begin{vmatrix} \frac{dp}{d\omega}, & \frac{dq}{d\omega}, & \dots \\ X, & P', & Q', & \dots \\ Y, & P'', & Q'', & \dots \\ Z, & P''', & Q''', & \dots \\ \vdots & \end{vmatrix},$$

where it will be remarked that the successive horizontal lines (after the first) of the determinant are the differential coefficients of $\Psi, p, q, \dots$ with respect to ξ , with respect to η , &c. In general, if ψ denote any function of $p, q, \dots$ these quantities being themselves functions of $\omega, \xi, \eta, \dots$, and $\xi, \eta, \dots$ containing ω ; also if Ψ be what ψ becomes when for $p, q, \dots$ we substitute their values in $\omega, \xi, \eta, \dots$; then we have identically

$$\begin{vmatrix} \frac{d\psi}{d\omega}, & \frac{dp}{d\omega}, & \frac{dq}{d\omega}, & \dots \\ X, & P, & Q, & \dots \\ Y, & P', & Q', & \dots \\ Z, & P'', & Q'', & \dots \\ \vdots & \end{vmatrix} = 0. \quad (?)$$

In the present case however, writing for shortness $\psi(p, q, \dots) = \psi$, this function ψ contains ω explicitly as well as implicitly through $p, q, \dots$. The formula is still true if for $\frac{d\psi}{d\omega}$ we substitute $\frac{d(\psi)}{d\omega} = \frac{d\psi}{d\omega} - \frac{d\Psi}{d\omega}$ on the second side, denoting a partial differential coefficient taken only so far as ω is explicitly contained in ψ . And considering $p, q, \dots$ as functions of $\omega, \xi, \eta, \dots$ ($\xi, \eta, \dots$ themselves functions of ω and of other variables which need not here be considered), ψ or Ψ vanishes identically, and we have $\frac{d(\psi)}{d\omega} = 0$. Hence, in the last formula, we have to write $-\frac{d\Psi}{d\omega}$ instead of $\frac{d\psi}{d\omega}$, and we thus derive

$$\begin{vmatrix} \frac{d\Psi}{d\omega} & P & Q & \dots \\ \cdot & P' & Q' & \dots \\ \cdot & P'' & Q'' & \dots \\ \vdots & \end{vmatrix} = \begin{vmatrix} \cdot, & \frac{dp}{d\omega}, & \frac{dq}{d\omega}, \dots \\ X, & P, & Q, \dots \\ Y, & P', & Q', \dots \\ Z, & P'', & Q'' \cdot \frac{d\Psi}{d\omega}, \dots \\ \vdots & \end{vmatrix}.$$

Footnote: This formula, or one equivalent to it, is given in Jacobi's memoir "De Determinantibus Functionalibus," Crelle, t. XXII. [1841] p. 319.

p. 441. whence D becomes

$$D = \begin{vmatrix} P & Q & \dots \\ P' & Q' & \dots \\ \vdots & \vdots & \end{vmatrix} \frac{d\Psi}{d\omega} S,$$

which is the value to be made use of in the equation

$$V = \int f\left(\frac{p}{\rho}, \frac{q}{\rho}, \dots\right) F(p, q, \dots) \frac{1}{\rho^{\mu+n}} D \, d\omega \, d\theta \, \dots$$

The principal use of the formula is where ψ is a homogeneous function of the second order of $p, q, \dots$. Thus, suppose

$$\psi = \frac{1}{2}(Ap^2 + Bq^2 + \dots + 2H pq + \dots),$$

also, for the sake of conformity to the usual notation in the theory of transformation of quadratic functions, writing $\alpha, \alpha', \dots, \beta, \beta', \dots$ instead of $P, P', \dots Q, Q', \dots$ and putting after the differentiations $\xi = 1$, we have

$$p = \alpha + \alpha' \eta + \alpha'' \zeta + \dots, \quad q = \beta + \beta' \eta + \beta'' \zeta + \dots,$$

values which we may assume to give rise to the equation

$$(Ap^2 + Bq^2 + \dots + 2H pq + \dots) = (1 - \eta^2 - \zeta^2 - \dots),$$

(where $\eta, \zeta, \dots$ are taken to be functions of θ , &c. such as to satisfy identically the equation $1 = \eta^2 + \zeta^2 + \dots$).

Hence, by a well-known property, if

$$\begin{vmatrix} A, & H, & \dots \\ H, & B, & \\ \vdots & \vdots & \end{vmatrix} = \kappa,$$

we have

$$\begin{vmatrix} \alpha, & \beta, & \dots \\ \alpha', & \beta', & \\ \vdots & \vdots & \end{vmatrix} = \sqrt{\frac{(-)^{n-1}}{\kappa}};$$

so that, observing that in the present case $X = 1$, and therefore

$$S = \begin{vmatrix} \eta, & \zeta, & \dots \\ \frac{d\eta}{d\theta}, & \frac{d\zeta}{d\theta}, & \\ \vdots & & \end{vmatrix},$$

we have

$$V = \int \frac{(-)^{\frac{1}{2}(n-1)}}{\kappa^{\frac{1}{2}}} f\left(\frac{p}{\rho}, \frac{q}{\rho}, \dots\right) F(p, q, \dots) \frac{1}{\rho^{\mu+n}} \frac{d\psi}{d\omega} S \, d\omega \, d\theta \, \dots$$

p. 442. The remainder of the process of integration may in many cases be effected by the method made use of by Jacobi in the memoir “De binis quibuslibet functionibus &c.” *Crelle*, t. XII. [1834] p. 1, viz. the coefficients α, α' , &c., β , &c., may in addition to the conditions which they are already supposed to satisfy, be so determined as to reduce any homogeneous function of $p, q, r, \dots$ entering into the integral to a form containing the squares only of the variables. This method is applied in the memoir in question to the integrals of n variables, analogous to those which give the attraction of an ellipsoid; and that directly without effecting an integration with respect to the radius vector. I proceed to show how the preceding investigations lead to Legendre’s integral, and how the method in question effects with the utmost simplicity the integration which Legendre accomplished by means of what Poisson has spoken of as inextricable calculations.

Consider in particular the formula

$$V = \int \frac{(x, y, \dots)^k \, dx \, dy \, \dots}{(x^2 + y^2 + \dots)^{i+\frac{1}{2}}},$$

the number of variables being as before n , and $(x, y, \dots)^k$ denoting a homogeneous function of the order k . The equation for the limits is assumed to be

$$l(x - a)^2 + m(y - b)^2 + \dots = k.$$

Assume

$$\psi(x, y, \dots) = \omega^2(x^2 + y^2 + \dots) + \delta(lx^2 + my^2 + \dots) - (lax + mby + \dots)^2,$$

(where $\delta = la^2 + mb^2 + \dots - k$, is taken to be positive); or more simply,

$$\psi(x, y, \dots) = (\omega^2 + l\delta - l^2a^2)x^2 + (\omega^2 + m\delta - m^2b^2)y^2 + \dots - 2lmabxy - \dots$$

Here $\mu = k + 2i - 3$. Also, putting for shortness

$$lap + mbq + \dots = \Lambda, \quad lp^2 + mq^2 + \dots = \Phi,$$

it is easy to obtain

$$f\left(\frac{p}{\rho}, \frac{q}{\rho}, \dots\right) = \frac{1}{k + 2i + n - 3} \frac{[-(\Lambda + \omega\rho)^{k+2i+n-3} + (\Lambda - \omega\rho)^{k+2i+n-3}]}{\Phi^{k+2i+n-3}}.$$

Also,

$$F(p, q, \dots) = \frac{(p, q, \dots)^k}{\rho^k}, \quad \frac{d\psi}{d\omega} = 2\omega\rho^2,$$

values which give

$$V = \frac{1}{k + 2i + n - 3} \int \frac{(-)^{\frac{1}{2}(n-1)}}{\kappa^{\frac{1}{2}}} \omega \rho^{n-1} \frac{[(\Lambda + \omega\rho)^{k+2i+n-3} - (\Lambda - \omega\rho)^{k+2i+n-3}]}{\Phi^{k+2i+n-3}} (p, q, \dots)^k S d\omega d\theta \dots.$$

p. 443. where it will be remembered that $p, q, \dots$ are linear functions (with constant terms) of $(n - 1)$ variables $\eta, \zeta, \dots$, these last mentioned quantities being themselves functions of $(n - 2)$ variables θ , &c. such that $1 - \eta^2 - \zeta^2 - \dots = 0$ identically. If besides we suppose that $\Phi = lp^2 + mq^2 + \dots$ reduces itself to the form $g\eta^2 + \dots$, we have, by the formula

[continuation of paper 74's transformation formula, applied here]

$$\left(1 - \frac{\lambda}{P}\right) \left(1 - \frac{\lambda}{Q}\right) = \frac{1}{\kappa} (\omega^2 + l\delta - l\lambda)(\omega^2 + m\delta - m\lambda) \dots,$$

which is true, whatever be the value of λ .

It seems difficult to proceed further with the general formula, and I shall suppose $n = 3$, $i = 0$, $k = 1$, $(x, y, \dots)^k = x$, or write

$$V = \int \frac{x dx dy dz}{(x^2 + y^2 + z^2)^{3/2}},$$

the equation of the limits being

$$l(x - a)^2 + m(y - b)^2 + n(z - c)^2 = k.$$

Here we may assume $\eta = \cos \phi$, $\zeta = \sin \phi$ (values which give $S = 1$). And we have

$$V = \int \omega^2 d\omega \int \frac{(\alpha + \alpha' \cos \phi + \alpha'' \sin \phi) d\phi}{lP^2 + mQ^2 + nR^2 \sin^2 \phi},$$

from $\phi = 0$ to $\phi = 2\pi$; or, what comes to the same thing,

$$V = \int \omega^2 d\omega \int \frac{\alpha d\phi}{lP^2 + mQ^2 + nR^2 \sin^2 \phi},$$

from $\phi = 0$ to $\phi = \frac{1}{2}\pi$. Hence

$$V = 4\pi la \int \frac{\omega^2 PQ R d\omega}{K \sqrt{[(B - mP)(C - nP) - F^2]}},$$

we have from the formulae of the paper before quoted,

$$\int \frac{d\phi}{lP^2 + mQ^2 + nR^2 \sin^2 \phi} = \frac{2\pi}{\sqrt{[(B - mP)(C - nP) - F^2]}},$$

B, C, F , being the coefficients of y^2, z^2, yz in $\psi(x, y, z)$, viz.

$$B = \omega^2 + m\delta - m^2b^2, \quad C = \omega^2 + n\delta - n^2c^2, \quad F = mnbc;$$

and consequently

$$V = 4\pi \int \frac{PQR d\omega}{\sqrt{[(B - mP)(C - nP) - F^2]}}.$$

p. 444. Also from the equation

$$\left(1 - \frac{l^2a^2}{\omega^2 + l\delta - lP}\right) \left(1 - \frac{m^2b^2}{\omega^2 + m\delta - mP}\right) \left(1 - \frac{n^2c^2}{\omega^2 + n\delta - nP}\right) = \frac{PQR}{K(P - Q)(P - R)}$$

differentiating with respect to λ , and writing $\lambda = P$,

$$-\frac{l^2a^2}{(\omega^2 + l\delta - lP)^2} \left(1 - \frac{m^2b^2}{\omega^2 + m\delta - mP}\right) \left(1 - \frac{n^2c^2}{\omega^2 + n\delta - nP}\right) = \frac{PQR}{(\omega^2 + l\delta - lP)(\omega^2 + m\delta - mP)(\omega^2 + n\delta - nP)} \cdot (\text{etc.})$$

or, as this may be written,

$$\frac{PQR}{K(P - Q)(P - R)} = \frac{1}{(\omega^2 + l\delta - lP)(\omega^2 + m\delta - mP)(\omega^2 + n\delta - nP)} \left\{ \frac{l^2a^2}{(\omega^2 + l\delta - lP)^2} + \frac{m^2b^2}{(\omega^2 + m\delta - mP)^2} + \frac{n^2c^2}{(\omega^2 + n\delta - nP)^2} \right\},$$

and from the values first written down, for B, C, F , we obtain $(B - mP)(C - nP) - F^2$

$$\begin{aligned} &= (\omega^2 + m\delta)(\omega^2 + n\delta) - m^2b^2(\omega^2 + n\delta) - n^2c^2(\omega^2 + m\delta) \\ &\quad - mP(\omega^2 + n\delta - n^2c^2) - nP(\omega^2 + m\delta - m^2b^2) + mnP^2 \\ &= (\omega^2 + m\delta - mP)(\omega^2 + n\delta - nP) - m^2b^2(\omega^2 + n\delta - nP) - n^2c^2(\omega^2 + m\delta - mP) \\ &= -\frac{\omega^2(\omega^2 + m\delta - mP)(\omega^2 + n\delta - nP)}{\omega^2 + l\delta - lP} \cdot \frac{l^2a^2}{1}, \end{aligned}$$

the last reduction being effected by means of the equation

$$1 - \frac{l^2a^2}{\omega^2 + l\delta - lP} - \frac{m^2b^2}{\omega^2 + m\delta - mP} - \frac{n^2c^2}{\omega^2 + n\delta - nP} = 0.$$

Hence

$$\frac{PQR\{(B - mP)(C - nP) - F^2\}^{1/2}}{K(P - Q)(P - R)} = \frac{la}{(\omega^2 + l\delta - lP)^{1/2}(\omega^2 + m\delta - mP)^{1/2}(\omega^2 + n\delta - nP)^{1/2}} \left\{ \frac{l^2a^2}{(\omega^2 + l\delta - lP)^2} + \frac{m^2b^2}{(\omega^2 + m\delta - mP)^2} + \frac{n^2c^2}{(\omega^2 + n\delta - nP)^2} \right\}.$$

Substituting this value, and multiplying out the fractions in the denominator,

$$V = 4\pi la \int \frac{\omega^2(\omega^2 + l\delta - lP)^{3/2}(\omega^2 + m\delta - mP)^{3/2}(\omega^2 + n\delta - nP)^{3/2} d\omega}{l^3a^2(\omega^2 + m\delta - mP)^2(\omega^2 + n\delta - nP)^2 + m^3b^2(\omega^2 + n\delta - nP)^2(\omega^2 + l\delta - lP)^2 + n^3c^2(\omega^2 + l\delta - lP)^2(\omega^2 + m\delta - mP)^2}$$

the reduction of which integral has been already treated of in the former part of this present memoir.

76. ON THE TRIPLE TANGENT PLANES OF SURFACES OF THE THIRD ORDER.

[From the *Cambridge and Dublin Mathematical Journal*, vol. iv. (1849), pp. 118–132.]

A surface of the third order contains in general a certain number of straight lines. Any plane through one of these lines intersects the surface in the line and in a conic, that is in a curve or system of the third order having two double points. Such a plane is therefore a double tangent plane of the surface, the double points (or points where the line and conic intersect) being the points of contact. By properly determining the plane, the conic will reduce itself to a pair of straight lines. Here the plane intersects the surface in three straight lines, that is in a curve or system of the third order having three double points, and the plane is therefore a triple tangent plane, the three double points or points of intersection of the lines taken two and two together being the points of contact. The number of lines and triple tangent planes is determined by means of a theorem very easily demonstrated, viz. that through each line there may be drawn five (and only five) triple tangent planes. Thus, considering any triple tangent plane, through each of the three lines

in this plane there may be drawn (in addition to the plane in question) four triple tangent planes: these twelve new planes give rise to twenty-four new lines upon the surface, making up with the former three lines, twenty-seven lines upon the surface. It is clear that there can be no lines upon the surface besides these twenty-seven; for since the three lines upon the triple tangent plane are the complete intersection of this plane with the surface, every other line upon the surface must meet the triple tangent plane in a point upon one of the three lines, and must therefore lie in a plane passing through one of these lines, such plane (since it meets the surface in two lines and therefore in a third line) being obviously a triple tangent plane. Hence the whole number of lines upon the surface is twenty-seven; and it immediately follows that the number of triple tangent planes is forty-five. The number of lines upon the surface may also be obtained by the following method, which has the advantage of not assuming *à priori* the existence of a line upon the surface.

p. 446. Imagine the cone having for its vertex a given point not upon the surface and circumscribed about the surface, every double tangent plane of the cone is also a double tangent plane of the surface, and therefore intersects the surface in a straight line (and a conic). And, conversely, if there be any line upon the surface, the plane through this line and the vertex of the cone will be a double tangent plane of the cone. Hence the number of double tangent planes of the cones is precisely that of the lines upon the surface. By the theorems in Mr Salmon's paper "On the degree of a surface reciprocal to a given one," *Journal*, vol. II. [1847] p. 65, the cone is of the sixth order and has no double lines and six cuspidal lines: hence by the formula in Plücker's "Theorie der algebraischen Curven," [1839] p. 211, stated so as to apply to cones instead of plane curves, viz. n being the order, x the number of double lines, y that of the cuspidal lines, u that of the double tangent planes, then

$$u = \frac{1}{2}n(n-2)(n^2-9) - (2x+3y)(n^2-n-6) + 2x(x-1) + 6xy + \frac{9}{2}y(y-1),$$

the number of double tangent planes is twenty-seven, which is therefore also the number of lines upon the surface.

Suppose the equation of one of the triple tangent planes to be $w = 0$, and let $x = 0$, $y = 0$, be the equation of any two triple tangent planes intersecting the plane $w = 0$ in two of the lines in which it meets the surface. Let $z = 0$ be the equation of a triple tangent plane meeting $w = 0$ in the remaining line in which it intersects the surface. The equation of the surface of the third order is in every case of the form $wP + kxyz = 0$, P being a function of the second order, but of the four different planes which the equation $z = 0$ may be supposed to represent, one of them such that the function P resolves itself into the product of a pair of factors, and for the remaining three this resolution into factors does not take place. This will be obvious from the sequel: at present I shall suppose that the plane $z = 0$ is of the latter class, or that $P = 0$ represents a proper surface of the second order. Since $x = 0$, $y = 0$, $z = 0$, are triple tangent planes of the surface, each of these planes must be a tangent plane of the surface of the second order $P = 0$, and this will be the case if we assume

$$P = x^2 + y^2 + z^2 + w^2,$$

and considering x, y, z and w as each of them implicitly containing an arbitrary constant, this is the most general function which satisfies the conditions in question.

We are thus led to the equation of the surface of the third order:

$$w(x^2 + y^2 + z^2 + w^2) + kxyz = 0.$$

p. 447. I have found that by expressing the parameter k in the particular form

$$k = -\frac{p^2 - \left(lmn - \frac{1}{lmn}\right)^2}{2\left(p - lmn - \frac{1}{lmn}\right)},$$

or, as this equation may be more conveniently written,

$$k = \frac{p^2 - \beta^2}{2(p - \alpha)}, \quad \alpha = lmn + \frac{1}{lmn}, \quad \beta = lmn - \frac{1}{lmn}, \quad (1)$$

the equations of all the planes are expressible in a rational form. These equations are in fact the following: [I have added, here and in the table p. 450, the reference numbers 12', 23', &c. constituting a different notation for the lines and planes.]

(w)	$w = 0,$	12'
(δ)	$lx + my + nz + w[1 + \frac{1}{k}(l - \frac{1}{l})(m - \frac{1}{m})(n - \frac{1}{n})] = 0,$	23'
(δ')	$\frac{x}{l} + \frac{y}{m} + \frac{z}{n} + w[1 + \frac{1}{k}(l - \frac{1}{l})(m - \frac{1}{m})(n - \frac{1}{n})] = 0,$	31'
(x)	$x = 0,$	12.34.56
(y)	$y = 0,$	42'
(z)	$z = 0,$	14'.23'
(ξ)	$x + \frac{1}{k}(m - \frac{1}{m})(n - \frac{1}{n})w = 0,$	1'2
(η)	$y + \frac{1}{k}(n - \frac{1}{n})(l - \frac{1}{l})w = 0,$	2'3
(ζ)	$z + \frac{1}{k}(l - \frac{1}{l})(m - \frac{1}{m})w = 0,$	3'1
(f)	$lx + \frac{y}{m} + \frac{z}{n} + w = 0,$	41'
(g)	$\frac{x}{l} + my + \frac{z}{n} + w = 0,$	34'
(h)	$\frac{x}{l} + \frac{y}{m} + nz + w = 0,$	13.24.56
(f̄)	$\frac{x}{l} + my + nz + w = 0,$	24'
(ḡ)	$lx + \frac{y}{m} + nz + w = 0,$	14.23.56
(h̄)	$lx + my + \frac{z}{n} + w = 0,$	43'
(ξ ₁)	$x + \frac{l(p-\alpha)+2mn}{p+\beta}(\frac{y}{m} + nz + w) = 0,$	12.35.46

[full table of forty-five planes continues; preserving structure with rational/irrational forms and the numerical labels 15', 51', 25', 26', 16, 16.24.35, etc.]

p. 449. In fact, representing the several functions on the left-hand side of these equations respectively by the letters placed opposite to them respectively, the function U is expressible in the sixteen forms following

$$\begin{aligned}
 U &= w\xi\xi_1 + k\xi\eta\zeta, \\
 &= w\eta\eta_1 + k\eta\zeta\xi, \\
 &= w\zeta\zeta_1 + k\zeta\xi\eta, \\
 &= wll_1 + kxyz, \\
 &= wmm_1 + kyzx, \\
 &= wnn_1 + kzxy, \\
 &= wpp_1 + k\xi yz, \\
 &= wqq_1 + k\eta zx, \\
 &= wrr_1 + k\xi xy, \\
 &= wpp_1 + k\xi yz, \\
 &= wqq_1 + k\eta zx, \\
 &= wrr_1 + k\zeta xy,
 \end{aligned}$$

(being the forms containing w , out of a complete system of one hundred and twenty different forms).

p. 450. The forty-five planes pass five and five through the twenty-seven lines in the following manner:

[A tabular array correlating each of the twenty-seven lines with the five planes through it, organised in three columns labelled by the lines a_i, b_i, c_i and using both alphabetic and numeric notations such as 12, 34, 56, 13', 35', etc.]

The twenty-seven lines lie three and three upon the forty-five planes in the following manner:

$$\begin{aligned}
 (w) \ a_1b_1c_1, \quad (\xi) \ a_4b_4c_4, \quad (l) \ a_7b_7c_7, \quad (p) \ a_2b_6c_8, \\
 (x) \ a_2b_5c_5, \quad (f) \ a_1b_4c_5, \quad (l) \ a_6b_9c_7, \quad (p_1) \ a_3b_8c_6, \\
 (\delta) \ a_3b_6c_8, \quad (h) \ c_5a_2b_1, \quad (n) \ c_4a_3b_5, \quad (r) \ c_8a_7b_2,
 \end{aligned}$$

p. 451.

$$\begin{array}{lll} (x) a_1 a_2 a_3, & (\xi) a_4 a_5 a_6, & (l) a_5 b_9 c_7, \quad (p_1) a_3 b_8 c_6, \\ (y) b_1 b_2 b_3, & (\eta) b_4 b_5 b_6, & (m_1) b_8 c_3 a_7, \quad (q_1) b_2 c_8 a_7, \\ (z) c_1 c_2 c_3, & (\zeta) c_4 c_5 c_6, & (n_1) c_4 a_5 b_3, \quad (r_1) c_3 a_7 b_8. \end{array}$$

The preceding method was the one that first occurred to me, and which appears to conduct most simply to the actual analytical expressions for the forty-five planes; but it is worth noticing that the relations between the lines and planes might have been obtained almost without algebraical developments, if we had supposed that P , instead of representing a proper surface of the second order, had represented a pair of planes. This would have conducted at once to one of the one hundred and twenty forms U , e.g. $U = w\theta\theta_1 + k\xi\eta\zeta$. Or changing the notation so as to include k in one of the linear functions, $U = ace - bdf$, and it is indeed obvious *à priori*, by merely reckoning the number of arbitrary constants, that any function of the third order can be put under this form. If we suppose $a = \mu b$ to be the equation of one of the triple tangent planes through the intersection of the planes a and b , the plane $a = \mu b$ meets the surface in the same lines in which it meets the hyperboloid $\mu ce - df = 0$, that is, the two lines in the plane are generating lines of different species, and consequently one of them meets the pair of lines cd and ef , and the other of them meets the pair of lines cf and de (where cd represents the line of intersection of the planes $c = 0$, $d = 0$, &c.). This suggests a notation for the lines in question, viz. each line may be represented by the three lines which it meets, or by the symbols $ab.cd.ef$ and $ab.cf.de$. Or observing that μ has three values, and that the same considerations apply *mutatis mutandis* to the planes through cb and ca , &c., the whole system of lines may be represented by the notation.

$$\begin{array}{lll} ab, & ad, & af, \\ cb, & cd, & cf, \\ eb, & ed, & ef, \\ (ab.cd.ef)_1, & (ab.cd.ef)_2, & (ab.cd.ef)_3, \\ (ad.cf.eb)_1, & (ad.cf.eb)_2, & (ad.cf.eb)_3, \\ (af.cb.ed)_1, & (af.cb.ed)_2, & (af.cb.ed)_3, \\ (ab.cf.ed)_1, & (ab.cf.ed)_2, & (ab.cf.ed)_3, \\ (ad.cb.ef)_1, & (ad.cb.ef)_2, & (ad.cb.ef)_3, \\ (af.cd.eb)_1, & (af.cd.eb)_2, & (af.cd.eb)_3. \end{array}$$

p. 452. where the last eighteen lines have been divided into two systems of nine each. The five planes through $(ab.cd.ef)_1$ may be considered as cutting the surface in

$$\begin{array}{l} ab; (ab.cf.ed)_1, \\ cd; (af.cd.eb)_1, \\ ef; (ad.cb.ef)_1, \\ (ad.cf.eb)_1; (af.cb.ed)_1, \\ (ad.cf.eb)_1; (af.cb.ed)_2, \end{array}$$

(which supposes however that the distinguishing suffixes 1, 2, 3, are added to the different planes according to a certain rule). And similarly for the lines in the planes through the other lines represented by symbols of the like form. The five planes through ab intersect the surface in the lines

$$\begin{array}{l} ab; \\ (ab.cd.ef)_1, \quad (ab.cf.ed)_1, \\ (ab.cd.ef)_2, \quad (ab.cf.ed)_2, \\ (ab.cd.ef)_3, \quad (ab.cf.ed)_3, \end{array}$$

and similarly for the planes through the other lines represented by symbols of a like form.

Observing that ξ, η, ζ , correspond to b, d, f , and w, θ, θ_1 , to a, c, e respectively, ab corresponds to the intersection of w and ξ , i.e. to a_1 , &c.; also $(ab.cd.ef)_1, (ab.cd.ef)_2, (ab.cd.ef)_3$ correspond to three lines

meeting a_1, b_1 , and c_1 , i.e. to a_4, a_5, a_6 , &c.; and the system of the twenty-seven lines as last written down corresponds to the system,

$$\begin{aligned} &a_1, b_1, c_1, \\ &a_2, b_2, c_2, \\ &a_3, b_3, c_3, \\ &a_4, a_5, a_6, \\ &b_4, b_5, b_6, \\ &c_4, c_5, c_6, \\ &a_7, a_8, a_9, \\ &b_7, b_8, b_9, \\ &c_7, c_8, c_9. \end{aligned}$$

The investigations last given are almost complete in themselves as the geometrical theory of the subject: there is however some difficulty in seeing *a priori* the nature of the correspondence between the planes which determines which are the planes which ought to be distinguished with the same one of the symbolic numbers, 1, 2, 3.

p. 453. There is great difficulty in conceiving the complete figure formed by the twenty-seven lines, indeed this can hardly I think be accomplished until a more perfect notation is discovered. In the mean time it is easy to find theorems which partially exhibit the properties of the system. For instance, any two lines, a_1, b_1 , which do not meet are intersected by five other lines, a_2, b_2, c_2, a_3, b_3 (no two of which meet). Any four of these last-mentioned lines are intersected by the lines a_1, b_1 and no other lines, but any three of them, e.g. a_2, a_3, b_3 , are intersected by the lines a_1, b_1 , and by some third line (in the case in question the line c_3). Or generally any three lines, no two of which meet, are intersected by three other lines, no two of which meet.

Again, the lines which do not meet any one of the lines a_1, a_2, a_3 , are $a_4, a_5, b_3, b_1, c_2, c_3$: these lines form a hexagon, the pairs of the opposite sides of which, a_2, b_3 ; a_3, c_2 ; b_2, c_1 , are met by the pairs a_1, b_2 ; c_2, a_1 and a_3, b_2 , respectively, viz. by pairs out of the system of three lines intersecting the system a_1, a_2, a_3 . And the lines a_1, a_2, a_3 may be considered as representing any three lines no two of which meet. Again, consider three lines in the same triple tangent plane, e.g. a_1, b_1, c_1 , and the hexahedron formed by any six triple tangent planes passing two and two through these lines, e.g. the planes $x, y, z, \xi, \eta, \zeta$. These planes contain (independently of the lines a_1, b_1, c_1) the twelve lines $a_2, a_3, a_4, a_5, b_2, b_3, b_4, b_5, c_1, c_3, c_4, c_5$. Consider three contiguous faces of the hexahedron, e.g. x, y, z , the lines in these planes, viz. $a_2, b_5, c_4, a_3, b_4, c_5$, form a hexagon the opposite sides of which intersect in a point, or in other words these six lines are generating lines of a hyperboloid. The same property holds for the systems ξ, η, ζ ; ξ, y, z ; ξ, η, z ; ξ, y, ζ . But for the system ξ, η, ζ , the six lines are a_4, b_4, c_4 , and a_5, b_5, c_5 , which form two triangles, and similarly for the systems ξ, y, z ; x, η, z ; and x, y, ζ ; so that the twelve lines form four hexagons (the opposite sides of which intersect) circumscribed round four of the angles of the hexahedron, and four pairs of triangles about the opposite four angles of the hexahedron. The number of such theorems might be multiplied indefinitely, and the number of different combinations of lines or planes to which each theorem applies is also very considerable.

Consider the four planes $x, f, \bar{x}, x$, and represent for a moment the equations of these planes by $x + Aw = 0$, $x + Bw = 0$, $x + Cw = 0$, $x + Dw = 0$, so that

$$A = 0, \quad B = \frac{1}{lmn} \left[\frac{1}{m} - n \right] \left[\frac{1}{n} - m \right], \quad C = \frac{l(p-\alpha)+2mn}{p+\beta}, \quad D = \frac{l(p-\alpha)+2mn}{p-\beta}.$$

By the assistance of

$$B - C = \frac{1}{lmn(p+\beta)} [ln(p-\alpha) + 2m] \cdot [lm(p-\alpha) + 2n],$$

it is easy to obtain

$$\frac{(A-C)(D-B)}{(A-D)(B-C)} = \frac{p-\beta}{p+\beta} \cdot \frac{[l(p-\alpha) + 2mn][m(p-\alpha) + 2n][n(p-\alpha) + 2l]}{[mn(p-\alpha) + 2l][nl(p-\alpha) + 2m][lm(p-\alpha) + 2n]},$$

p. 454. which remains unaltered for cyclical permutations of l, m, n , i.e. the anharmonic ratio of $x, f, \bar{x}, x$ is the same as that of $y, \eta, \bar{y}, y$ or $z, \zeta, \bar{z}, z$; there is of course no correspondence of x to y or ξ to η , &c., the

correspondence is by the general properties of anharmonic ratios, a correspondence of the system $x, f, \bar{x}, x$, to any one of the systems $(y, \eta, \bar{y}, y)$, or $(\eta, y, \bar{y}, y)$, or $(y, y, \bar{y}, \eta)$, or $(y, y, \bar{\eta}, y)$, indifferently. The theorems may be stated generally as follows: “Considering two lines in the same triple tangent plane, the remaining triple tangent planes through these two lines respectively are homologous systems.”

Suppose the surface of the third order intersected by an arbitrary plane. The curve of intersection is of course one of the third order, and the positions upon this curve of six of the points in which it is intersected may be arbitrarily assumed. Let these points be the points in which the plane is intersected by the lines $a_1, b_1, a_2, b_2, c_2, a_3$; or as we may term them, the points $a_1, b_1, a_2, b_2, c_2, a_3$.^(?) The point c_1 is of course the point in which the line a_1b_1 intersects the curve. The straight lines $a_1b_4c_4, b_1c_4a_4, c_1a_4b_4$, and $a_2b_4c_5, b_2c_4a_5, c_2a_4b_5$, show that c_4 and b_4 are the points in which a_2b_4 and c_2c_4 intersect the curve, and then b_5 and c_5 are determined as the intersections of a_2c_4, a_3b_4 with the curve. The intersection of the lines b_5 and a_3c_5 (which is known to be a point upon the curve by the theorem, every curve of the third order passing through eight of the points of intersection of two curves of the third order passes through the ninth point of intersection) is the point a_5 . The systems $a_4, b_4, c_4; a_3, b_3, c_3; a_5, b_5, c_5$, determine the conjugate system $a_6, b_6, c_6; a_7, b_7, c_7; a_8, b_8, c_8$; by reason of the straight lines $a_1a_4a_5, b_1b_4b_5, c_1c_4c_5; a_1a_2a_3, b_1b_2b_3, c_1c_2c_3; a_1a_6a_7, b_1b_6b_7, c_1c_6c_7$, viz. a_6 is the point where a_1a_4 intersects the curve, &c. The relations of the systems $(a_4, b_4, c_4; a_5, b_5, c_5), (a_6, b_6, c_6; a_7, b_7, c_7), (a_8, b_8, c_8; a_9, b_9, c_9)$ to the system $a_1, b_1, c_1; a_2, b_2, c_2; a_3, b_3, c_3$ are precisely identical. It is only necessary to show how the points $a_2, b_2, c_2; a_3, b_3, c_3$ of the latter system are determined by means of one of the former systems, suppose the system $a_6, b_6, c_6; a_7, b_7, c_7$; and to discover a compendious statement of the relation between the two systems. The points $a_1, b_1, c_1; a_2, b_2, c_2; a_3, b_3, c_3; a_6, b_6, c_6; a_7, b_7, c_7$, are a system of fifteen points lying on the fifteen straight lines

$(a_1b_1c_1, a_2b_2c_2, a_3b_3c_3, a_1a_2a_3, b_1b_2b_3, c_1c_2c_3, a_1a_6a_7, b_1b_6b_7, c_1c_6c_7, a_2b_7c_6, b_2a_7c_6, c_2a_7b_6, a_3b_6c_7, b_3a_6c_7, c_3a_6b_7),$

viz. the nine points $a_1, b_1, c_1; a_2, b_2, c_2; a_3, b_3, c_3$ are the points of intersection of the three lines $a_1b_1c_1, a_2b_2c_2, a_3b_3c_3$ with the three lines $a_1a_2a_3, b_1b_2b_3, c_1c_2c_3$, and the remaining six points form a hexagon $a_6b_7c_6a_7b_6c_7$, of which the diagonals a_6a_7, b_6b_7, c_6c_7 pass through the points a_1, b_1, c_1 , respectively, the alternate sides a_6b_7, c_6a_7 , and b_6c_7 pass through the points c_2, b_2, a_2 , respectively, and the remaining alternate sides b_7c_6, a_7b_6 , and c_7a_6 pass through the three points a_3, b_3, c_3 respectively. The fifteen points of such a system do not necessarily lie upon a curve of the third order, as will presently be seen: in the actual case however where all the points lie upon a given curve of the third order, and the points $a_1, b_1, c_1; a_6, b_6, c_6; a_7, b_7, c_7$ are known, $a_2, b_2, c_2; a_3, b_3, c_3$ are the intersections of the curve with $b_7c_6, c_6a_7, a_7b_6, b_6c_7, c_7a_6, a_6b_7$ respectively, and the fact

Footnote: In general, the point in which any line upon the surface intersects the plane in question may be represented by the symbol of the line, and the line in which any triple tangent plane intersects the plane in question may be represented by the symbol of the triple tangent plane: thus, a_1, b_1, c_1 are points in the line $a_1b_1c_1$, or in the line $w, \mathcal{E}c$.

p. 455. of the existence of the lines $a_1b_1c_1, a_2b_2c_2, a_3b_3c_3, a_1a_2a_3, b_1b_2b_3, c_1c_2c_3$ is an immediate consequence of the theorem quoted above with respect to curves of the third order—a theorem from which the entire system of relations between the twenty-seven points on the curve might have been deduced *à priori*. But returning to the system of fifteen points, suppose the lines $a_1b_1c_1, a_2b_2c_2, a_3b_3c_3$, and $a_1a_2a_3, b_1b_2b_3$, and also the point a_6 to be given arbitrarily. The point a_7 lies on the line a_1a_6 , suppose its position upon this line to be arbitrarily assumed (in which case, since the ten points $a_1, a_2, a_3, b_1, b_2, b_3, c_1, c_2, c_3$, are sufficient to determine a curve of the third order, there is no curve of the third order through these points and the point a_6). If the points b_6, c_6, b_7, c_7 can be so determined that the sides of the quadrilateral $b_6b_7c_6c_7$, viz. $b_6b_7, b_7c_6, c_6c_7, c_7b_6$ pass through the points b_1, a_2, c_1, a_3 , respectively, while the angles b_6, b_7, c_6, c_7 lie upon the lines a_7c_3, a_6c_3, a_7b_3 and a_6b_3 respectively, the required conditions will be satisfied by the fifteen points in question; and the solution of this problem is known. I have not ascertained whether in the case of an arbitrary position as above of the point a_7 , it is possible to determine a complete system of twenty-seven points lying three and three upon forty-five lines in the same manner as the twenty-seven points upon the curve of the third order; but it appears probable that this is the case, and to determine whether it be so or not, presents itself as an interesting problem for investigation.

Suppose that the intersecting plane coincides with one of the triple tangent planes. Here we have a system of twenty-four points, lying eight and eight in three lines; the twenty-four points lie also three and

three in thirty-two lines, which last-mentioned lines therefore pass four and four through the twenty-four points. If we represent by $a, b, c, d, a', b', c', d'$ and $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}, \mathbf{a}', \mathbf{b}', \mathbf{c}', \mathbf{d}'$, the eight points, and eight points

p. 456. which lie upon two of the three lines (the order being determinate), the systems of four lines which intersect in the eight points of the third line are

$$\begin{aligned} (a\mathbf{a}, b\mathbf{b}, c\mathbf{c}, d\mathbf{d}), & \quad (a'\mathbf{a}', b'\mathbf{b}', c'\mathbf{c}', d'\mathbf{d}'), \\ (a\mathbf{b}', b\mathbf{a}', c'\mathbf{d}, d'\mathbf{c}), & \quad (a'\mathbf{b}, b'\mathbf{a}, c\mathbf{d}', d\mathbf{c}'), \\ (a\mathbf{c}', c\mathbf{a}', d'\mathbf{b}, b'\mathbf{d}), & \quad (a'\mathbf{c}, c'\mathbf{a}, b\mathbf{d}', d\mathbf{b}'), \\ (a\mathbf{d}', d\mathbf{a}', b'\mathbf{c}, c'\mathbf{b}), & \quad (a'\mathbf{d}, d'\mathbf{a}, b\mathbf{c}', c\mathbf{b}'); \end{aligned}$$

the principle of symmetry made use of in this notation (which however represents the actual symmetry of the system very imperfectly) being obviously entirely different from that of the case of an arbitrary intersecting plane. The transition case where the intersecting plane passes through one of the lines upon the surface (and is thus a double tangent plane) would be worth examining. It should be remarked that the preceding theory is very materially modified when the surface of the third order has one or more conical points; and in the case of a double line (for which the surface becomes a ruled surface) the theory entirely ceases to be applicable. I may mention in conclusion that the whole subject of this memoir was developed in a correspondence with Mr Salmon, and in particular, that I am indebted to him for the determination of the number of lines upon the surface and for the investigations connected with the representation of the twenty-seven lines by means of the letters a, c, e, b, d, f , as developed above.

77. ON THE ORDER OF CERTAIN SYSTEMS OF ALGEBRAICAL EQUATIONS.

[From the *Cambridge and Dublin Mathematical Journal*, vol. IV. (1849), pp. 132–137.⁽¹⁾]

Suppose the variables $x, y, \dots$ so connected that any one of the ratios $x : y : z, \dots$, or, more generally, any determinate function of these ratios, depends on an equation of the μ^{th} order. The variables $x, y, z, \dots$ are said to form a system of the μ^{th} order.

In the case of two variables x, y , supposing that these are connected by an equation $U = 0$ (U being a homogeneous function of the order μ) the variables form a system of the μ^{th} order; and, conversely, whenever the variables form a system of the μ^{th} order, they are connected by an equation of the above form.

In the case of a greater number of variables, the question is one of much greater difficulty. Thus with three variables x, y, z ; if μ be resolvable into the factors μ', μ'' , then, supposing the variables to be connected by the equations $U = 0, V = 0$, U and V being homogeneous functions of the orders μ', μ'' , respectively, they will it is true form a system of the μ^{th} order, but the converse proposition does not hold: for instance, if μ is a prime number, the only mode of forming a system of the μ^{th} order would on the above principle be to assume $\mu' = \mu, \mu'' = 1$, that is to suppose the variables connected by an equation of the μ^{th} order and a linear equation; but this is far from being the most general method of obtaining such a system. In fact, systems not belonging to the class in question may be obtained by the introduction of subsidiary

Footnote ⁽¹⁾: This memoir was intended to appear at the same time with Mr Salmon's "Note on a Result of Elimination," *Journal*, vol. III. p. 169 with which it is very much connected.

p. 458. variables to be eliminated: the simplest example is the following: suppose a, b, a', b', a'', b'' to be linear functions (without constant terms) of x, y, z , and write

$$\begin{cases} a\xi + b\eta = 0, \\ a'\xi + b'\eta = 0, \\ a''\xi + b''\eta = 0; \end{cases}$$

equations from which, by the elimination of ξ, η , two relations may be obtained between the variables x, y, z .

Suppose, however, from these three equations x, y, z are first eliminated: the ratio $\xi : \eta$ will evidently be determined by a cubic equation; and assuming $\xi : \eta$ to be equal to one of the roots of this, any two of the three equations may be considered as implying the third; and will likewise determine linearly the

ratios $x : y : z$. Hence any determinate function of these ratios depends on a cubic equation only, or the system is one of the third order. But the order of the system may be obtained by means of the equations resulting from the elimination of ξ, η ; and since this will explain the following more general example (in which the corresponding process is the only one which readily offers itself), it will be convenient to deduce the preceding result in this manner. Thus, performing the elimination, we have

$$L = (a'b'' - a''b') = 0, \quad L' = (a''b - ab'') = 0, \quad L'' = (ab' - a'b) = 0.$$

Here the equations $L = 0$, $L' = 0$, $L'' = 0$, are each of them of the second order, and any two of them may be considered as implying the third. For we have identically,

$$aL + a'L' + a''L'' = 0,$$

so that $L = 0$, $L' = 0$, gives $a''L'' = 0$, or $L'' = 0$. Nevertheless the system is imperfectly represented by means of two equations only. For instance, $L = 0$, $L' = 0$ do, of themselves, represent a system which is really of the fourth order. In fact, these equations are satisfied by $a'' = 0$, $b'' = 0$, (which is to be considered as forming a system of the first order), but these values do not satisfy the remaining equation $L'' = 0$. In other words, the equations $L = 0$, $L' = 0$ contain an extraneous system of the first order, and which is seen to be extraneous by means of the last equation L'' : the system required is the system of the third order which is common to the three equations $L = 0$, $L' = 0$, $L'' = 0$.

Suppose, more generally, that x, y, z are connected by $p + 1$ equations, involving p variables $\xi, \eta, \zeta, \dots$,

$$a\xi + b\eta + c\zeta + \dots = 0,$$

Footnote: Also $bL + b'L' + b''L'' = 0$: but since by the elimination of L'' , taking into account the actual values of L and L' , we obtain an identical equation, these two relations may be considered as equivalent to a single one.

p. 459. or what comes to the same by the equations (equivalent to two independent relations)

$$\begin{vmatrix} a, & a', & a'', & \dots & a^{(p)} \\ b, & b', & b'', & \dots & b^{(p)} \end{vmatrix} = 0,$$

(where the number of horizontal rows is p). Consider x, y, z , as connected by the two equations

$$a\xi + b\eta + c\zeta + \dots = 0, \quad a'\xi + b'\eta + c'\zeta + \dots = 0;$$

these form a system of the order p^2 , but they involve the extraneous system

$$\begin{vmatrix} a'', & \dots & a^{(p)} \\ b'', & \dots & b^{(p)} \end{vmatrix} = 0.$$

Suppose $\phi(p)$ is the order of the system in question, then the order of this last system is $\phi(p-1)$ and hence $\phi(p) = p^2 - \phi(p-1)$: observing that $\phi(2) = 3$, this gives directly $\phi(p) = \frac{1}{2}p(p+1)$. Hence the order of the system is $\frac{1}{2}p(p+1)$.

Suppose x, y, z , connected by equations of the form $U = 0$, $V = 0$, $W = 0$; U, V, W being linear in x, y, z , and homogeneous functions of the orders m, n, p respectively in ξ, η . By eliminating x, y, z , the ratio $\xi : \eta$ will be determined by an equation of the order $m + n + p$; and since when this is known the ratios $x : y : z$ are linearly determinable, we have $m + n + p$ for the order μ of the system.

Thus, if $m = n = p = 2$, selecting the particular system

$$\begin{aligned} a\xi^2 + 2b\xi\eta + c\eta^2 &= 0, \\ b\xi^2 + 2c\xi\eta + d\eta^2 &= 0, \\ c\xi^2 + 2d\xi\eta + e\eta^2 &= 0, \end{aligned}$$

it is possible in this case to obtain two resulting equations of the orders two and three respectively, and which consequently constitute the system of the sixth order without containing any extraneous system. In fact, from the identical equation

$$\begin{aligned} &e\xi^2 \cdot (a\xi^2 + 2b\xi\eta + c\eta^2) \\ &- (4d\xi + 2e\eta)(b\xi^2 + 2c\xi\eta + d\eta^2) \\ &+ (3c\xi + 2d\eta)(c\xi^2 + 2d\xi\eta + e\eta^2) = (ae - 4bd + 3c^2)\xi^2, \end{aligned}$$

p. 460. and

$$\begin{aligned} & (ce - d^2)(a\xi^2 + 2b\xi\eta + c\eta^2) \\ & + (cd - be)(b\xi^2 + 2c\xi\eta + d\eta^2) \\ & + (bd - c^2)(c\xi^2 + 2d\xi\eta + e\eta^2) = (ace - ad^2 - b^2e - c^3 + 2bcd)\xi^2, \end{aligned}$$

we deduce

$$\begin{cases} ae - 4bd + 3c^2 = 0, \\ ace - ad^2 - b^2e - c^3 + 2bcd = 0; \end{cases}$$

which form the system in question, and may for shortness be represented by $I = 0, J = 0$.

The three equations in ξ, η may be considered as expressing that

$$\begin{aligned} a\xi^3 + 3b\xi^2\eta + 3c\xi\eta^2 + d\eta^3 &= 0, \\ b\xi^3 + 3c\xi^2\eta + 3d\xi\eta^2 + e\eta^3 &= 0, \end{aligned}$$

have a pair of equal roots in common; in other words, that it is possible identically to satisfy

$$(A\xi + B\eta)(a\xi^3 + 3b\xi^2\eta + 3c\xi\eta^2 + d\eta^3) + (A'\xi + B'\eta)(b\xi^3 + 3c\xi^2\eta + 3d\xi\eta^2 + e\eta^3) = 0.$$

Equating to zero the separate terms of this equation, and eliminating A, B, A', B' , we obtain

$$\begin{vmatrix} \cdot & a, & 3b, & 3c, & d \\ \cdot & b, & 3c, & 3d, & e \\ a, & 3b, & 3c, & d, & \cdot \\ b, & 3c, & 3d, & e, & \cdot \end{vmatrix} = 0.$$

It is not at first sight obvious what connection these equations have with the two, $I = 0, J = 0$, but by actual expansion they reduce themselves to the following five,

$$\begin{aligned} 3[2(ce - d^2)I - 3eJ] &= 0, \\ 3[(be - cd)I - 3dJ] &= 0, \\ [-(ae + 2bd - 3c^2)I + 9cJ] &= 0, \\ 3[(ad - bc)I - 3bJ] &= 0, \\ 3[2(ac - b^2)I - 3aJ] &= 0; \end{aligned}$$

which are satisfied by $I = 0, J = 0$. By the theorem above given, the equations are to be considered as forming a system of the tenth order; the system must therefore be considered as composed of the system $I = 0, J = 0$, and of a system of the fourth order. The system of the fourth order may be written in the form

$$2(ac - b^2) : ad - bc : ae + 2bd - 3c^2 : be - cd : 2(ce - d^2) = a : b : 3c : d : e;$$

p. 461. but to justify this, it must be shown first that these equations reduce themselves to two independent equations; and next that system is really one of the fourth order. We may remark in the first place, that if

$$u = a\xi^4 + 4b\xi^3\eta + 6c\xi^2\eta^2 + 4d\xi\eta^3 + e\eta^4$$

is a perfect square, the coefficients will be proportional to those of $\frac{d^2u}{d\xi^2 d\eta^2} - \left(\frac{d^2u}{d\xi d\eta}\right)^2$, ⁽¹⁾. Thus the conditions requisite in order that u may be a perfect square, are given by the system

$$\begin{aligned} 2(ac - b^2) : (ad - bc) : ae + 2bd - 3c^2 : be - cd : 2(ce - d^2) \\ = a : b : 3c : d : e, \end{aligned}$$

or these equations are equivalent to two independent equations only (this may be easily verified *à posteriori*); and by writing $3J$ in the form

$$e(ac - b^2) - 2d(ad - bc) + c(ae + 2bd - 3c^2) - 2b(be - cd) + a(ce - d^2),$$

the remaining equations of the complete system (3) are immediately deduced; thus the latter system contains only two independent equations. (The preceding reasoning shows that the system (3) expresses the conditions in order that the equations

$$a\xi^3 + 3b\xi^2\eta + 3c\xi\eta^2 + d\eta^3 = 0, \quad b\xi^3 + 3c\xi^2\eta + 3d\xi\eta^2 + e\eta^3 = 0,$$

may have a pair of unequal roots in common: we have already seen that the equations $I = 0$, $J = 0$ represent the conditions in order that these two equations may have a pair of equal roots in common.) Finally, to verify *à posteriori* the fact of the system (3) being one only of the fourth order, we may, as Mr Salmon has done in the memoir above referred to, represent the system by the two equations

$$a(ce - d^2) - e(ac - b^2) = 0, \quad e(ad - bc) - 2b(ce - d^2) = 0,$$

that is, by

$$ad^2 - eb^2 = 0, \quad 2bd^2 - 3bce + ae^2 = 0.$$

These equations contain the extraneous system ($a = 0$, $b = 0$) and the extraneous systems ($b = 0$, $d = 0$) and ($d = 0$, $e = 0$), each of which last, as Mr Salmon has remarked from geometrical considerations, counts double, or the system is one of the 4th order only.

Footnote ⁽¹⁾: More generally whatever be the order of u , if u contain a square factor, this square factor may easily be shown to occur in $\frac{d^2u}{d\xi^2} \frac{d^2u}{d\eta^2} - \left(\frac{d^2u}{d\xi d\eta}\right)^2$.

78. NOTE ON THE MOTION OF ROTATION OF A SOLID OF REVOLUTION.

[From the *Cambridge and Dublin Mathematical Journal*, vol. IV. (1849), pp. 268–270.]

Using the notation employed in my former papers on the subject of rotation (*Cambridge Math. Journal*, vol. III. pp. 224–232, [6]; and *Cambridge and Dublin Math. Journal*, vol. I. pp. 167–264, [37]), suppose $B = A$, then r is constant, equal to n suppose; and writing

$$\frac{(A - C)n}{A} = \nu, \quad \text{or } C = \left(1 - \frac{\nu}{n}\right)A;$$

also putting

$$\theta = \nu t + \gamma,$$

(where γ is an arbitrary constant) the values of p, q, r are easily seen to be given by the equations

$$\begin{cases} p = M \sin \theta, \\ q = M \cos \theta, \\ r = n, \end{cases}$$

(where M is arbitrary). And consequently

$$\begin{aligned} h &= A[M^2 + n(n - \nu)], \\ k^2 &= A^2[M^2 + (n - \nu)^2]. \end{aligned}$$

Also, since $a^2 + b^2 + c^2 = k^2$, we may write

$$\begin{aligned} a &= -k \sin i \cos j, \\ b &= k \cos i \cos j, \\ c &= k \sin j; \end{aligned}$$

k having the value above given, and the angles i, j being arbitrary.

p. 463. From the equations (12) and (15) in the second of the papers quoted, we deduce

$$\begin{aligned} 2\nu &= k\{k + A(n - \nu) \sin j + MA \cos j \cos(\theta + i)\}, \\ \Phi &= k\{n \sin j + M \cos j \cos(\theta + i)\}, \\ \nabla &= -\nu k M A \cos j \sin(\theta + i), \end{aligned}$$

(values which verify as they should do the equation (19)). Hence, from the equation (27), writing $\frac{2\nu}{\Phi} dt = -\frac{1}{\nu} d\theta$, we have

$$2 \tan^{-1} \frac{\Omega}{k} = \delta + \frac{1}{\nu} \int d\theta \cdot \frac{k + kn \sin j + kM \cos j \cos(\theta + i)}{k + A(n - \nu) \sin j + MA \cos j \cos(\theta + i)}.$$

This is easily integrated; but the only case which appears likely to give a simple result is when the quantity under the integral sign is constant, or

$$A(k + kn \sin j) = k[k + A(n - \nu) \sin j],$$

that is,

$$Ah - k^2 + Ak\nu \sin j = 0;$$

whence

$$A(n - \nu) + k \sin j = 0 : \\ \sin j = \frac{A(n - \nu)}{k}, \quad \cos j = \frac{AM}{k}, \quad \text{or } \tan j = \frac{-(n - \nu)}{M} = -\frac{C}{A} \frac{n}{M}.$$

Observing that $\frac{1}{2}\pi - j$ is the inclination of the axis of z to the normal to the invariable plane, this equation shows that the supposition above is not any restriction upon the generality of the motion, but amounts only to supposing that the axis of z (which is a line fixed in space) is taken upon the surface of a certain right cone having for its axis the perpendicular to the invariable plane. Resuming the solution of the problem, we have

$$2 \tan^{-1} \frac{\Omega}{k} = \delta + \frac{k}{\nu A} \theta,$$

which may also be written under the form

$$2 \tan^{-1} \frac{\Omega}{k} = \delta_1 + \frac{kt}{A},$$

(where $\delta_1 = \delta + \frac{k\gamma}{\nu A}$). And hence

$$\Omega = k \tan \frac{1}{2} \left(\delta_1 + \frac{kt}{A} \right) = k \tan \frac{1}{2} \psi,$$

where

$$\psi = \frac{1}{2} \left(\delta_1 + \frac{kt}{A} \right).$$

p. 464. Substituting these values,

$$a = -MA \sin i, \quad b = MA \cos i, \quad c = -A(n - \nu),$$

$$w = A^2 M^2 \{1 + \cos(\theta + i)\} = 2M^2 A^2 \cos^2 \frac{1}{2}(\theta + i);$$

and substituting in the equations (14) the values of λ, μ, ν reduce themselves to

$$\lambda = \frac{1}{MA \cos \frac{1}{2}(\theta + i)} [k \tan \frac{1}{2} \psi \sin \frac{1}{2}(\theta - i) - A(n - \nu) \cos \frac{1}{2}(\theta - i)],$$

$$\mu = \frac{1}{MA \cos \frac{1}{2}(\theta + i)} [k \tan \frac{1}{2} \psi \cos \frac{1}{2}(\theta - i) + A(n - \nu) \sin \frac{1}{2}(\theta - i)],$$

$$\nu = \tan \frac{1}{2}(\theta + i);$$

where, recapitulating, $\theta = \nu t + \gamma$, $2\psi = \delta + \frac{kt}{A} \delta_1$.

I may notice, in connexion with the problem of rotation, a memoir, “Specimen Inaugurale de motu gyrationis corporis rigidi &c.,” by A. S. Rueb (Utrecht, 1834), which contains some very interesting developments of the ordinary solution of the problem, by means of the theory of elliptic functions.

79. ON A SYSTEM OF EQUATIONS CONNECTED WITH Malfatti's Problem, AND ON ANOTHER ALGEBRAICAL SYSTEM.

[From the *Cambridge and Dublin Mathematical Journal*, vol. iv. (1849), pp. 270–275.]

Consider the equations

$$\begin{aligned} by^2 + cz^2 + 2fyz &= \theta\theta'a(bc - f^2), \\ cz^2 + ax^2 + 2gzx &= \theta\theta'b(ca - g^2), \\ ax^2 + by^2 + 2hxy &= \theta\theta'c(ab - h^2); \end{aligned}$$

or, as they may be more conveniently written,

$$\begin{aligned} by^2 + cz^2 + 2fyz &= \theta\theta'a\mathfrak{A}, \\ cz^2 + ax^2 + 2gzx &= \theta\theta'b\mathfrak{B}, \\ ax^2 + by^2 + 2hxy &= \theta\theta'c\mathfrak{C}. \end{aligned}$$

The second and third equations give

$$(g^2 - h^2)x^2 - b^2y^2 + c^2z^2 + 2cg\mathfrak{C}zx - 2bh\mathfrak{B}xy = 0,$$

hence $\{(g\sqrt{\mathfrak{C}} - h\sqrt{\mathfrak{B}})x - b\sqrt{\mathfrak{B}}y + c\sqrt{\mathfrak{C}}z\}^2 - \mathfrak{A}(-bgy + chz)^2 = 0$, and consequently

$$(g\sqrt{\mathfrak{C}} - h\sqrt{\mathfrak{B}})x - b\sqrt{\mathfrak{B}}(g\sqrt{\mathfrak{A}} + h\sqrt{\mathfrak{B}})y + c\sqrt{\mathfrak{C}}(g\sqrt{\mathfrak{A}} + h\sqrt{\mathfrak{C}})z = 0;$$

dividing this by $g\sqrt{\mathfrak{A}} + h\sqrt{\mathfrak{B}}$, and writing down the system of equations to which the equation thus obtained belongs,

$$\begin{aligned} (g\sqrt{\mathfrak{C}} - h\sqrt{\mathfrak{B}})x - b\sqrt{\mathfrak{B}}y + c\sqrt{\mathfrak{C}}z &= 0, \\ a\sqrt{\mathfrak{A}}x + (h\sqrt{\mathfrak{A}} - f\sqrt{\mathfrak{C}})y - c\sqrt{\mathfrak{C}}z &= 0, \\ -a\sqrt{\mathfrak{A}}x + b\sqrt{\mathfrak{B}}y + (f\sqrt{\mathfrak{B}} - g\sqrt{\mathfrak{A}})z &= 0. \end{aligned}$$

p. 466. Hence also

$$\begin{aligned} (a\sqrt{\mathfrak{A}} + b\sqrt{\mathfrak{B}} - f\sqrt{\mathfrak{C}})y - (g\sqrt{\mathfrak{A}} - f\sqrt{\mathfrak{B}} + c\sqrt{\mathfrak{C}})z &= 0, \\ (a\sqrt{\mathfrak{A}} - h\sqrt{\mathfrak{B}} + b\sqrt{\mathfrak{C}})x - (-f\sqrt{\mathfrak{A}} + b\sqrt{\mathfrak{B}} + f\sqrt{\mathfrak{C}})y &= 0; \end{aligned}$$

these equations may be written

$$\begin{aligned} \frac{1}{k}\{\mathfrak{F}\sqrt{\mathfrak{A}} - \mathfrak{G}\sqrt{\mathfrak{B}} - \mathfrak{H}\sqrt{\mathfrak{C}} + \sqrt{(\mathfrak{A}\mathfrak{B}\mathfrak{C})}\}[\{\mathfrak{G} + \sqrt{(\mathfrak{C}\mathfrak{A})}\}y - \{\mathfrak{H} + \sqrt{(\mathfrak{A}\mathfrak{B})}\}z] &= 0, \\ \frac{1}{k}\{-\mathfrak{F}\sqrt{\mathfrak{A}} + \mathfrak{G}\sqrt{\mathfrak{B}} - \mathfrak{H}\sqrt{\mathfrak{C}} + \sqrt{(\mathfrak{A}\mathfrak{B}\mathfrak{C})}\}[\{\mathfrak{H} + \sqrt{(\mathfrak{A}\mathfrak{B})}\}z - \{\mathfrak{F} + \sqrt{(\mathfrak{B}\mathfrak{C})}\}x] &= 0, \\ \frac{1}{k}\{-\mathfrak{F}\sqrt{\mathfrak{A}} - \mathfrak{G}\sqrt{\mathfrak{B}} + \mathfrak{H}\sqrt{\mathfrak{C}} + \sqrt{(\mathfrak{A}\mathfrak{B}\mathfrak{C})}\}[\{\mathfrak{F} + \sqrt{(\mathfrak{B}\mathfrak{C})}\}x - \{\mathfrak{G} + \sqrt{(\mathfrak{C}\mathfrak{A})}\}y] &= 0; \end{aligned}$$

where, as usual,

$$\begin{aligned} \mathfrak{F} &= gh - af, \quad \mathfrak{G} = hf - bg, \quad \mathfrak{H} = fg - ch, \\ K &= abc - af^2 - bg^2 - ch^2 + 2fgh; \end{aligned}$$

(in fact the coefficient of y in the first equation is

$$\frac{1}{k}\{(\mathfrak{F}\mathfrak{G} - \mathfrak{C}\mathfrak{H})\sqrt{\mathfrak{A}} + (\mathfrak{A}\mathfrak{C} - \mathfrak{G}^2)\sqrt{\mathfrak{B}} - (\mathfrak{G}\mathfrak{H} - \mathfrak{A}\mathfrak{F})\sqrt{\mathfrak{C}}\} = -\mathfrak{F}\sqrt{\mathfrak{A}} + \mathfrak{G}\sqrt{\mathfrak{B}} - f\sqrt{\mathfrak{C}},$$

as it should be, and similarly for the coefficients of the remaining terms). We have therefore

$$\{\mathfrak{F} + \sqrt{(\mathfrak{B}\mathfrak{C})}\}x = \{\mathfrak{G} + \sqrt{(\mathfrak{C}\mathfrak{A})}\}y = \{\mathfrak{H} + \sqrt{(\mathfrak{A}\mathfrak{B})}\}z;$$

or, what comes to the same thing,

$$\begin{aligned} yz &= \frac{1}{a}s\{\mathfrak{F} + \sqrt{(\mathfrak{B}\mathfrak{C})}\}, \\ zx &= \frac{1}{b}s\{\mathfrak{G} + \sqrt{(\mathfrak{C}\mathfrak{A})}\}, \\ xy &= \frac{1}{c}s\{\mathfrak{H} + \sqrt{(\mathfrak{A}\mathfrak{B})}\}. \end{aligned}$$

Now

$$\begin{aligned} a\{\mathfrak{G} + \sqrt{(\mathfrak{C}\mathfrak{A})}\}\{\mathfrak{H} + \sqrt{(\mathfrak{A}\mathfrak{B})}\} &= \{\mathfrak{F} + \sqrt{(\mathfrak{B}\mathfrak{C})}\}\{abc - fgh + f\sqrt{(\mathfrak{B}\mathfrak{C})} - g\sqrt{(\mathfrak{C}\mathfrak{A})} - h\sqrt{(\mathfrak{A}\mathfrak{B})}\}, \\ b\{\mathfrak{H} + \sqrt{(\mathfrak{A}\mathfrak{B})}\}\{\mathfrak{F} + \sqrt{(\mathfrak{B}\mathfrak{C})}\} &= \{\mathfrak{G} + \sqrt{(\mathfrak{C}\mathfrak{A})}\}\{abc - fgh - f\sqrt{(\mathfrak{B}\mathfrak{C})} + g\sqrt{(\mathfrak{C}\mathfrak{A})} - h\sqrt{(\mathfrak{A}\mathfrak{B})}\}, \\ c\{\mathfrak{F} + \sqrt{(\mathfrak{B}\mathfrak{C})}\}\{\mathfrak{G} + \sqrt{(\mathfrak{C}\mathfrak{A})}\} &= \{\mathfrak{H} + \sqrt{(\mathfrak{A}\mathfrak{B})}\}\{abc - fgh - f\sqrt{(\mathfrak{B}\mathfrak{C})} - g\sqrt{(\mathfrak{C}\mathfrak{A})} + h\sqrt{(\mathfrak{A}\mathfrak{B})}\}, \end{aligned}$$

(as readily appears by writing the first of these equations under the form

$$a\{\mathfrak{G} + \sqrt{(\mathfrak{C}\mathfrak{A})}\}\{\mathfrak{H} + \sqrt{(\mathfrak{A}\mathfrak{B})}\} = \{\mathfrak{F} + \sqrt{(\mathfrak{B}\mathfrak{C})}\}\{abc - fgh + f\sqrt{(\mathfrak{B}\mathfrak{C})} - g\sqrt{(\mathfrak{C}\mathfrak{A})} - h\sqrt{(\mathfrak{A}\mathfrak{B})}\},$$

and comparing the rational term and the coefficients of $\sqrt{(\mathfrak{B}\mathfrak{C})}$, $\sqrt{(\mathfrak{C}\mathfrak{A})}$, $\sqrt{(\mathfrak{A}\mathfrak{B})}$.

p. 467. Hence, observing the values of yz, zx, xy , we find

$$\begin{aligned} x^2 &= \frac{s}{2a}\{abc - fgh + f\sqrt{(\mathfrak{B}\mathfrak{C})} - g\sqrt{(\mathfrak{C}\mathfrak{A})} - h\sqrt{(\mathfrak{A}\mathfrak{B})}\}, \\ y^2 &= \frac{s}{2b}\{abc - fgh - f\sqrt{(\mathfrak{B}\mathfrak{C})} + g\sqrt{(\mathfrak{C}\mathfrak{A})} - h\sqrt{(\mathfrak{A}\mathfrak{B})}\}, \\ z^2 &= \frac{s}{2c}\{abc - fgh - f\sqrt{(\mathfrak{B}\mathfrak{C})} - g\sqrt{(\mathfrak{C}\mathfrak{A})} + h\sqrt{(\mathfrak{A}\mathfrak{B})}\}. \end{aligned}$$

Hence, forming the value of any one of the functions $by^2 + cz^2 + 2fyz$, $cz^2 + ax^2 + 2gxy$, $ax^2 + by^2 + 2hxy$, we obtain $s = \theta\theta'$, or we have

$$\begin{aligned} x^2 &= \frac{\theta\theta'}{2a}\{abc - fgh + f\sqrt{(\mathfrak{B}\mathfrak{C})} - g\sqrt{(\mathfrak{C}\mathfrak{A})} - h\sqrt{(\mathfrak{A}\mathfrak{B})}\}, \\ y^2 &= \frac{\theta\theta'}{2b}\{abc - fgh - f\sqrt{(\mathfrak{B}\mathfrak{C})} + g\sqrt{(\mathfrak{C}\mathfrak{A})} - h\sqrt{(\mathfrak{A}\mathfrak{B})}\}, \\ z^2 &= \frac{\theta\theta'}{2c}\{abc - fgh - f\sqrt{(\mathfrak{B}\mathfrak{C})} - g\sqrt{(\mathfrak{C}\mathfrak{A})} + h\sqrt{(\mathfrak{A}\mathfrak{B})}\}, \\ yz &= \frac{\theta\theta'}{2a}\{\mathfrak{F} + \sqrt{(\mathfrak{B}\mathfrak{C})}\}, \\ zx &= \frac{\theta\theta'}{2b}\{\mathfrak{G} + \sqrt{(\mathfrak{C}\mathfrak{A})}\}, \\ xy &= \frac{\theta\theta'}{2c}\{\mathfrak{H} + \sqrt{(\mathfrak{A}\mathfrak{B})}\}. \end{aligned}$$

It may be remarked that the equations

$$by^2 + cz^2 + 2fyz = L, \quad cz^2 + ax^2 + 2gzx = M, \quad ax^2 + by^2 + 2hxy = N,$$

in which the coefficients are supposed to be such that the functions $M(a^2x^2 + b^2y^2 + 2h^2xy) = N(a^2x^2 + b^2y^2 + 2f^2yz) = L(c^2z^2 + a^2x^2 + 2g^2zx)$, are each of them decomposable into linear factors, may always be reduced to a system of equations similar to those which have just been solved.

Suppose

$$f = g = h = \frac{1}{\theta'} = \sqrt{\frac{abc}{a+b+c}} = r,$$

and writing $\sqrt{X}, \sqrt{Y}, \sqrt{Z}$ instead of x, y, z . The equations to be solved become

$$\begin{aligned} bY + cZ + 2r\sqrt{YZ} &= (b+c)r, \\ cZ + aX + 2r\sqrt{ZX} &= (c+a)r, \\ aX + bY + 2r\sqrt{XY} &= (a+b)r. \end{aligned}$$

p. 468. where $r^2 = \frac{abc}{a+b+c}$, and the solution is

$$\begin{aligned} x &= \frac{r}{2a}[a+b+c-r+\sqrt{(r^2+a^2)}-\sqrt{(r^2+b^2)}-\sqrt{(r^2+c^2)}], \\ y &= \frac{r}{2b}[a+b+c-r-\sqrt{(r^2+a^2)}+\sqrt{(r^2+b^2)}-\sqrt{(r^2+c^2)}], \\ z &= \frac{r}{2c}[a+b+c-r-\sqrt{(r^2+a^2)}-\sqrt{(r^2+b^2)}+\sqrt{(r^2+c^2)}], \\ \sqrt{(yz)} &= \frac{1}{2}\{r-a+\sqrt{(r^2+a^2)}\}, \\ \sqrt{(zx)} &= \frac{1}{2}\{r-b+\sqrt{(r^2+b^2)}\}, \\ \sqrt{(xy)} &= \frac{1}{2}\{r-c+\sqrt{(r^2+c^2)}\}, \end{aligned}$$

a system of formulae which contain the solution of the problem “In a given triangle to inscribe three circles such that each circle touches the remaining two circles and also two sides of the triangle.” In fact, if r denote the radius of the inscribed circle, and a, b, c the distances of the angles of the triangle from the points where the sides are touched by the inscribed circle (quantities which it is well known satisfy the condition $r^2 = \frac{abc}{a+b+c}$), also if x, y, z denote the radii of the required circles, there is no difficulty in

obtaining by the determination of x, y, z , the above system of equations. The problem in question was first proposed and solved by an Italian geometer named Malfatti, and has been called after him Malfatti's problem. His solution, dated 1803, and published in the 10th volume of the Transactions of the Italian Academy of Sciences, appears to have consisted in showing that the values first found for the radii of the three circles satisfy the equations given above, without any indication of the process of obtaining the expressions for these radii. Further information as to the history of the problem may be found in the memoir "Das Malfattische Problem neu gelöst von C. Adams," Winterthur, 1846.

In connexion with the preceding investigations may be considered the problem of determining l and m from the equations

$$\begin{aligned} B(1 + \theta)^2 - 2H(1 + \theta)m + (A + 1)m^2 &= 0, \\ A(m + \theta)^2 - 2H(m + \theta)l + (B + 1)l^2 &= 0; \end{aligned}$$

which express that the function

$$\theta^2 U + (lx + my)^2, \quad (U = Ax^2 + 2Hxy + By^2),$$

has for one of its factors a factor of $U + x^2$, and for the other of its factors a factor of $U + y^2$. There is no difficulty in solving the above equations; and if we write

$$K = AB - H^2, \quad \omega_1 = \sqrt{(-K - B)}, \quad \omega_2 = \sqrt{(-K - A)},$$

p. 469. the result is easily shown to be

$$l : m : \theta = A(B + H + \omega_1) : B(A + H + \omega_2) : (H + \omega_1)(H + \omega_2) - AB.$$

But the problem may be considered as the problem for two variables, analogous to that of determining the conic having a double contact with a given conic, and touching three conics each of them having a double contact with the given conic; and in this point of view I was led to the following solution. If we assume

$$Bl - Hm = u, \quad -Hl + Am = v,$$

or, what is the same thing,

$$Kl = Au + Hv, \quad Km = Hu + Bv,$$

then putting

$$\nabla = Au^2 + 2Huv + Bv^2,$$

the two equations become after some reduction

$$\begin{aligned} (u - K\theta)^2 &= -\varpi_1^2 \left(K\theta^2 + \frac{1}{K}\nabla \right), \\ (v - K\theta)^2 &= -\varpi_2^2 \left(K\theta^2 + \frac{1}{K}\nabla \right). \end{aligned}$$

Hence, writing

$$\begin{aligned} K\theta^2 + \frac{1}{K}\nabla &= -s^2, \quad \text{we have} \\ u &= K\theta + s\varpi_1, \quad v = K\theta + s\varpi_2, \quad \nabla = K^2\theta^2 + Ks^2 = 0; \end{aligned}$$

and substituting these values of u, v in the last equation,

$$A(K\theta + s\varpi_1)^2 + 2H(K\theta + s\varpi_1)(K\theta + s\varpi_2) + B(K\theta + s\varpi_2)^2 + K(K\theta^2 + Ks^2) = 0,$$

or reducing,

$$K\theta^2(A + 2H + B + 1) + 2K\theta s\{(A + H)\varpi_1 + (H + B)\varpi_2\} + s^2\{A\varpi_1^2 + 2H\varpi_1\varpi_2 + B\varpi_2^2 + K\} = 0;$$

whence

$$\begin{aligned} &\{K\theta(A + 2H + B + 1) + s[(A + H)\varpi_1 + (H + B)\varpi_2]\}^2 \\ &= s^2\{[(A + H)\varpi_1 + (H + B)\varpi_2]^2 - (A + 2H + B + 1)(A\varpi_1^2 + 2H\varpi_1\varpi_2 + B\varpi_2^2 + K)\} \\ &= s^2\{-(A + H)\varpi_1^2 - (H + B)\varpi_1^2 - (A + B + 1)(A\varpi_1^2 + 2H\varpi_1\varpi_2 + B\varpi_2^2) \\ &\quad - K(A + 2H + B + 1) - 2(A + H)(B + H)\varpi_1\varpi_2 + 2H\varpi_1\varpi_2 \\ &\quad + B^2\varpi_2^2 + 2AH\varpi_1^2 + AB\varpi_2^2 + 2AH\varpi_1\varpi_2\} \\ &= s^2\{2H\varpi_1\varpi_2 + K - H\}^2; \end{aligned}$$

p. 470. and therefore

$$K\theta(A + 2H + B + 1) + s\{(A + H)\varpi_1 + (H + B)\varpi_2\} = s\{2\varpi_1\varpi_2 + K - H\},$$

giving

$$s = \frac{K\theta(A + 2H + B + 1)}{\varpi_1\varpi_2 - (A + H)\varpi_1 - (H + B)\varpi_2 + K - H}.$$

But

$$Kl - Au + Hv = (A + H)\varpi_1 + (A + H)\varpi_2 + (A + H)s, \quad Km = (H + B)\varpi_1 + (H + B)\varpi_2 + (H + B)s,$$

and substituting the above value of s , we obtain, after some simple reductions,

$$l : m : \theta = (A + H)\varpi_1 + (A - H)\varpi_2 \cdot (\varpi_1\varpi_2 + K - H) : (\varpi_1 + \varpi_2) K\theta : \dots$$

[*continuation*] a result which presents itself in a very different form from the one previously obtained; if, however, the terms of this proportion be multiplied by the factor

$$\frac{H\varpi_1\varpi_2 - K\varpi_1 - K\varpi_2 + AB - KH}{(A + 2H + B + 1)K},$$

they become (as they ought to do) identical with those of the former proportion, and the identical relations to which this process gives rise are not without interest.

80. SUR QUELQUES TRANSMUTATIONS DES LIGNES COURBES.

[From the *Journal de Mathématiques Pures et Appliquées* (Liouville), tom. XIV. (1849), pp. 40–46.]

Je suppose qu'une courbe quelconque soit représentée par une équation homogène entre trois variables x, y, z , et je me propose d'examiner (en ne faisant attention qu'aux droites et aux coniques) ce que signifient les transmutations

$$\begin{aligned} \text{I. } & \xi = \sqrt{x}, \quad \eta = \sqrt{y}, \quad \zeta = \sqrt{z}, \\ \text{II. } & \xi = x^2, \quad \eta = y^2, \quad \zeta = z^2, \\ \text{III. } & \xi = \frac{1}{x}, \quad \eta = \frac{1}{y}, \quad \zeta = \frac{1}{z}, \end{aligned}$$

ces quantités étant prises dans chaque cas pour de nouvelles coordonnées. Les théorèmes auxquels on se trouve conduit comprennent les théorèmes qu'obtient M. William Roberts par sa méthode générale de transmutation, en écrivant $n = \frac{1}{2}$, $n = 2$. (*Voir* sa Note sur ce sujet, t. XIII. [1848] de ce Journal, page 209.)⁽¹⁾

I. Soit

$$\xi = \sqrt{x}, \quad \eta = \sqrt{y}, \quad \zeta = \sqrt{z}.$$

Je supposerai ici et partout dans ce mémoire que PQR soit le triangle formé par les droites

$$x = 0, \quad y = 0, \quad z = 0.$$

Footnote ⁽¹⁾: M. William Roberts a bien voulu me parler l'été passé à Dublin, avant que cette Note eût paru, des théorèmes qui devaient s'y trouver: cela me suggira, et je lui communiquai, peu de jours après, la deuxième et la troisième de mes méthodes de transmutation. On verra dans la suite que ce sont la première et la deuxième de ces méthodes qui correspondent aux cas $n = \frac{1}{2}, n = 2$, respectivement, mais que la troisième est aussi très étroitement liée avec le cas $n = \frac{1}{2}$.

p. 472. Cela posé, en considérant les nouvelles coordonnées ξ, η, ζ , on voit tout de suite qu'un système d'équations telles que

$$\xi : \eta : \zeta = \alpha : \beta : \gamma$$

correspond à un point; qu'une équation linéaire quelconque

$$A\xi + B\eta + C\zeta = 0$$

correspond à une conique qui touche les trois côtés du triangle PQR (ou, si l'on veut, inscrite à ce triangle); et qu'une équation quelconque du second ordre

$$a\xi^2 + b\eta^2 + c\zeta^2 + 2f\eta\zeta + 2g\zeta\xi + 2h\xi\eta = 0$$

correspond à une courbe du quatrième ordre qui touche les trois côtés du triangle PQR , chaque côté deux fois. Et réciproquement, de telles coniques et de telles courbes du quatrième ordre peuvent toujours se représenter par une équation linéaire ou par une équation du second ordre entre les coordonnées ξ, η, ζ .

On déduit de là cette propriété générale:

“Tout théorème descriptif qui se rapporte à des points, à des droites, et à des coniques, conduit à un théorème qui se rapporte d'une manière analogue à des points, à des coniques qui touchent chacune trois droites fixes, et à des courbes du quatrième ordre qui touchent chacune ces trois droites fixes, deux fois chaque droite.”

En supposant les coefficients g, h se réduisant à zéro, l'équation

$$a\xi^2 + b\eta^2 + c\zeta^2 + 2f\eta\zeta = 0$$

est celle d'une conique tangente aux deux droites PQ, PR , et la courbe du quatrième ordre se réduit à deux coniques égales et superposées l'une à l'autre. De là:

“Tout théorème descriptif qui se rapporte à des points, à des droites, et à une conique, conduit à un théorème qui se rapporte d'une manière analogue à des points, à des coniques qui touchent chacune trois droites fixes, deux fois chaque droite.”

En particulier, en supposant que les deux droites fixes que la conique touche soient les deux droites tangentes à cette conique qui passent par un des foyers, et que la troisième droite fixe soit à l'infini:

“Tout théorème descriptif qui se rapporte à des droites, et à une conique, conduit à un théorème qui se rapporte d'une manière analogue à des points, à des paraboles qui ont pour foyer commun un point fixe, et à une conique qui a aussi ce point fixe pour un de ses foyers.”

II. Soit à présent

$$\xi = x^2, \quad \eta = y^2, \quad \zeta = z^2.$$

Un système d'équations telles que

$$\xi : \eta : \zeta = \alpha : \beta : \gamma$$

p. 473. correspond à quatre points qui formeront ce que l'on peut nommer un *système symétrique par rapport aux points P, Q, R* , ou tout simplement, un *système symétrique* (on voit sans peine que les quatre points P, Q, R , et le même système symétrique sont, tels, que le quadrilatère formé par ces quatre points a pour points de concours des diagonales et des côtés opposés les points P, Q, R ou simplement les points P, Q, R , est indépendant du choix du système symétrique; on entendra dans la suite par les mots *système symétrique par rapport aux P, Q, R* , ou simplement *courbe symétrique par rapport aux points P, Q, R* , ou simplement *courbe symétrique*, toute courbe lieu d'un système symétrique. Cela posé, il y a une infinité de coniques symétriques; savoir, toute conique par rapport à laquelle les points P, Q, R sont des points conjugués (cela veut dire que la conique en est une pour le polaire de l'un des trois autres). On peut d'ailleurs nommer une telle courbe *courbe symétrique par rapport aux points P, Q, R* , ou simplement *courbe symétrique*, toute conique lieu c'est qu'il y a une infinité de coniques symétriques; savoir, toute conique par rapport à laquelle les points P, Q, R sont des points conjugués. Remarquons qu'une courbe du quatrième ordre peut se réduire à un système de deux coniques telles, que les points de chaque système symétrique sont, tels, que le quadrilatère de ces points a pour polaire la droite menée par les deux autres). De cela, considérons une courbe symétrique du quatrième ordre passant à laquelle l'ensemble des deux coniques soient partagées deux à deux par les deux coniques. En effet, considérons telle, que, par rapport à cette conique, le point P ait pour polaire la droite QR . Il est facile de construire une seule conique telle, que l'ensemble des trois coniques sont des coniques symétriques. Pour ce qui regarde une courbe symétrique du quatrième ordre, on a aussi un système de deux coniques symétriques telle, que, par rapport à cette conique, l'un des trois points fixes a pour polaire la droite QR . Il est facile de construire une seule conique telle, que l'ensemble des trois coniques sont des coniques symétriques. Pour ce qui regarde une courbe symétrique du quatrième ordre, on a aussi un système de deux coniques symétriques. Nous dirons que les deux coniques sont des coniques supplémentaires.

En revenant à notre but actuel, une équation linéaire quelconque

$$A\xi + B\eta + C\zeta = 0$$

correspond à une conique symétrique. Une équation du second ordre quelconque

$$a\xi^2 + b\eta^2 + c\zeta^2 + 2f\eta\zeta + 2g\zeta\xi + 2h\xi\eta = 0$$

correspond à une courbe symétrique du quatrième ordre. Et réciproquement, toute conique symétrique, ou toute courbe symétrique du quatrième ordre, peut se représenter par une équation linéaire, ou du second ordre, avec les coordonnées ξ, η, ζ . De là cette propriété générale:

“Tout théorème descriptif qui se rapporte à des points, à des droites, et à des coniques, conduit à un théorème qui se rapporte d’une manière analogue à des systèmes symétriques (par rapport à trois points fixes), à des coniques symétriques, et à des courbes symétriques du quatrième ordre.”

Dans ce théorème on peut, si l’on veut, considérer l’un des points d’un système symétrique comme représentant le système, et substituer aux mots *systèmes symétriques* (par rapport à trois points fixes), le mot *points*.

En supposant que dans la courbe du quatrième ordre on ait à la fois

$$\omega - g^2 = 0, \quad ah - h^2 = 0,$$

il est facile de voir que la courbe du quatrième ordre se réduit à deux coniques supplémentaires.

p. 474. En effet, ces conditions étant remplies, en rétablissant les valeurs de ξ, η, ζ , on obtient une équation qui se divise en deux équations, telles que

$$ax^2 + \beta y^2 + \gamma z^2 - \delta yz = 0, \quad ax^2 + \beta y^2 + \gamma z^2 + \delta yz = 0,$$

qui appartiennent, comme on le voit sans peine, à une paire de coniques supplémentaires.

De là, en remarquant qu’il est permis de faire abstraction de l’une de ces coniques:

“Tout théorème qui se rapporte à des points, à des droites, et à une conique, conduit à un théorème qui se rapporte d’une manière analogue à des points, à des coniques symétriques (par rapport à trois points fixes), et à une conique symétrique également (par rapport à ces trois points fixes), qui ne passe pas par les deux autres points fixes.”

Supposons en particulier que les deux points fixes dont nous venons de parler soient les trois points en l’infini, et considérons un troisième point fixe au-dehors. Il y a une infinité de coniques telles que les trois points fixes sont des points conjugués; ces coniques sont évidemment quelconques avec les droites imaginaires asymptotes de la conique symétrique fixée, conduisant ainsi un faisceau harmonique, et de là les asymptotes de la conique dont il s’agit seront un angle droit, ou, autrement dit, cette conique sera une hyperbole équilatère. De là enfin:

“Tout théorème descriptif qui se rapporte à des droites, à des points, et à une conique, conduit à un théorème qui se rapporte d’une manière analogue à des points, à des hyperboles équilatères et concentriques, et à une conique concentrique avec ces hyperboles.”

III. Soit enfin

$$\xi = \frac{1}{x}, \quad \eta = \frac{1}{y}, \quad \zeta = \frac{1}{z}.$$

Cette supposition conduit à l’une des méthodes de transmutation données par M. Steiner parmi les observations générales qui forment la conclusion de son ouvrage intitulé: *Systematische Entwicklung* &c., [Berlin, 1832], méthode qu’obtient M. Steiner au moyen de la théorie de l’hyperboloïde gauche. Je la reproduis ici tant pour ce faire voir la théorie analytique qui en cause de son analogie avec les méthodes I. et II.

Il est évident d’abord qu’un système d’équations telles que

$$\xi : \eta : \zeta = \alpha : \beta : \gamma$$

correspond à un point; de même, une équation linéaire quelconque, telle que

$$A\xi + B\eta + C\zeta = 0,$$

correspond à une conique qui passe par les trois points P, Q, R , et une équation quelconque du second ordre, telle, que

$$a\xi^2 + b\eta^2 + c\zeta^2 + 2f\eta\zeta + 2g\zeta\xi + 2h\xi\eta = 0,$$

p. 475. correspond à une courbe du quatrième ordre qui a les trois points P, Q, R pour points doubles. Réciproquement, de telles coniques et de telles courbes du quatrième ordre peuvent toujours se représenter par des équations linéaires et par des équations quelconques du second ordre, telles que

$$A\xi + B\eta + C\zeta = 0.$$

De là:

“Tout théorème descriptif qui se rapporte à des points, à des droites, et à des coniques, conduit à un théorème qui se rapporte d’une manière analogue à des points, à des coniques qui passent par trois points fixes, et à des courbes du quatrième ordre qui ont ces trois points fixes pour points doubles.”

En supposant que l’on ait à la fois $a = 0$, $h = 0$, la courbe du quatrième ordre se réduit à une conique qui passe par les deux points Q, R (ou, si l’on veut, au système formé de cette conique et des droites PQ, PR). De là:

“Tout théorème descriptif qui se rapporte à des points, à des droites, et à une conique, conduit à un théorème qui se rapporte d’une manière analogue à des points, à des coniques qui passent par trois points fixes, et à une conique qui passe par deux de ces points fixes.”

On ne peut pas particulariser ce théorème de manière à obtenir des théorèmes intéressants. Mais, en prenant les polaires réciproques des deux théorèmes qui viennent d’être obtenus, on a ces propriétés nouvelles:

“Tout théorème descriptif qui se rapporte à des points, à des droites, et à des coniques, conduit à un théorème qui se rapporte d’une manière analogue à des droites, à des coniques qui touchent chacune trois droites fixes, et à des courbes de la quatrième classe qui touchent chacune ces mêmes droites fixes, deux fois chaque droite;”

et:

“Tout théorème qui se rapporte à des droites, à des points, et à une conique, conduit à un théorème qui se rapporte d’une manière analogue à des droites, à des coniques qui touchent chacune trois droites fixes, et à une conique qui touche deux de ces droites fixes.”

De là, comme dans la méthode I.:

“Tout théorème qui se rapporte à des droites, à un point, à une conique, conduit à un théorème qui se rapporte d’une manière analogue à des droites, à des paraboles qui ont pour foyer commun un point fixe, et à une conique qui a ce même point fixe pour un de ses foyers.”

théorème que l’on doit comparer avec le troisième théorème que donne la méthode I.

Pour compléter cette théorie, nous aurions dû ajouter les deux théorèmes polaires réciproques du premier et du deuxième théorème que donne la méthode I.; mais cela ne fait aucun la moindre peine. Quant au premier et au deuxième théorème que donne la méthode II., les polaires réciproques de ces deux théorèmes ne conduisent à rien de nouveau.

81. ADDITION AU MÉMOIRE SUR QUELQUES TRANSMUTATIONS DES LIGNES COURBES.

[From the *Journal de Mathématiques Pures et Appliquées* (Liouville), tom. xv. (1850), pp. 351–356:
continued from t. XIV. pp. 46, 80.]

Je me propose de résumer ici la théorie des courbes du quatrième ordre, auxquelles donne lieu la première de mes méthodes de transmutation appliquée à une conique quelconque.

L’équation d’une telle courbe est de la forme

$$ax + by + cz + 2f\sqrt{yz} + 2g\sqrt{zx} + 2h\sqrt{xy} = 0,$$

et nous avons déjà vu que cette courbe a pour tangentes doubles les trois droites $x = 0$, $y = 0$, $z = 0$ (droites que nous avons représentées par QR, RP, PQ). Pour trouver les autres propriétés de la courbe, mettons l’équation sous la forme

$$\left(a - \frac{gh}{f}\right)x + \left(b - \frac{hf}{g}\right)y + \left(c - \frac{fg}{h}\right)z + \left(\sqrt{\frac{gh}{f}}\sqrt{x} + \sqrt{\frac{hf}{g}}\sqrt{y} + \sqrt{\frac{fg}{h}}\sqrt{z}\right)^2 = 0;$$

puis, en écrivant, pour plus de simplicité, $\frac{fg}{gh}, \frac{gh}{hf}, \frac{fg}{hf}$ au lieu de x, y, z , cette équation devient

$$\left(\frac{af}{gh} - 1\right)x + \left(\frac{bg}{hf} - 1\right)y + \left(\frac{ch}{fg} - 1\right)z + (\sqrt{x} + \sqrt{y} + \sqrt{z})^2 = 0;$$

et de là, en mettant

$$\left(\frac{af}{gh} - 1\right)x + \left(\frac{bg}{hf} - 1\right)y + \left(\frac{ch}{fg} - 1\right)z = -w,$$

p. 477. l'équation de la courbe prend cette forme très simple

$$\sqrt{x} + \sqrt{y} + \sqrt{z} + \sqrt{w} = 0,$$

en se souvenant toujours que les quantités x, y, z, w satisfont à l'équation linéaire qui vient d'être donnée. Je représenterai dans la suite cette équation linéaire par

$$ax + \beta y + \gamma z + \delta w = 0.$$

Il est évident que la droite $w = 0$, de même que les droites QR, RP, PQ , est tangente double de la courbe. De plus, ces quatre droites sont les six systèmes complets des tangentes doubles, car la courbe a, comme nous allons le voir, trois points doubles; en effet, la forme rationnelle de l'équation est

$$(x^2 + y^2 + z^2 + w^2 - 2yz - 2zx - 2xy - 2yw - 2zw)^2 - 64xyzw = 0,$$

et, au moyen de l'identité

$$(x + y)^2(z + w)^2 - 16xyzw = (x - y)^2(z + w)^2 + (z - w)^2(x + y)^2 - (x - y)^2(z - w)^2,$$

cette équation rationnelle se transforme en

$$[(x - y)^2 - (z - w)^2]^2 - 4\{(x - y + z - w)^2 + (z - w)^2(x + y)\}\{(x - y)^2 - (z - w)^2\} = 0,$$

laquelle fait voir que le point $(x = y, z = w)$ est un point double; de là aussi les points $(z = w, x = y)$, $(x = w, y = z)$ sont des points doubles. Remarquons en passant qu'on supposant que les coefficients $\alpha, \beta, \gamma, \delta$ restent indéterminés, les droites $x = y, z = w$ sont également, et ces deux droites passent par les deux points $(y = 0, z = 0)$, $(x = 0, w = 0)$, c'est-à-dire respectivement, en droites $y = 0, z = 0$ communs, les droites $y = z, x = w$, et la troisième au moyen des points $(x = 0, w = 0)$, $(z = w, x = y)$, $(x = 0, w = 0)$ et la troisième au moyen des points $(y = 0, z = 0)$, $(x = w, y = z)$, et la troisième au moyen des points $(z = 0, w = 0)$, $(x = y, z = w)$ sont les trois droites doubles qui déterminent ces coupures nécessairement dans un même point. Cela revient au théorème suivant:

“Les trois points doubles d'une courbe du quatrième ordre dont les six tangentes doubles sont les côtés d'un quadrangle dont les côtés passent par les angles du quadrilatère formé par les tangentes doubles de la courbe.”

Cette propriété des courbes du quatrième ordre dont il s'agit (je veux dire celles d'avoir trois points doubles) aurait dû faire partie du théorème général donné auparavant pour cette première méthode de transmutation.

En supposant que la conique à transmuter passe par le point P , on aura

$$a + \delta = 0,$$

et il suit de là que le point double $(x = w, y = z)$, identique dans ce cas avec le point P , se change en plus de points de rebroussement, ou en même temps que les droites PQ, PR ne sont plus que des tangentes doubles propres dites, mais se réduisent à des

p. 478. tangentes simples qui passent par le point de rebroussement. Ajoutons que la tangente au point de rebroussement est la droite $x = w$.

En supposant que la conique à transmuter passe à la fois par les deux points P et Q , nous aurons

$$a + \delta = 0, \quad \beta + \delta = 0.$$

Ici les deux points doubles $(x = w, y = z)$, $(y = w, z = x)$, identiques dans ce cas avec les points P, Q , deviennent des points de rebroussement, les droites QR, RP, PQ , et la droite w , ne sont plus des tangentes doubles propres dites, les droites RP, PQ ne sont plus que des tangentes simples qui passent par les deux points de rebroussement respectivement, et la droite PQ est la droite menée par les deux points de rebroussement. Ajoutons que les tangentes aux deux points de rebroussement respectivement sont les droites $x = w$ et $y = w$.

On sait qu'une courbe quelconque dont s'envisage comme conique qui passe par deux points fixes à l'infini, contient ce qu'on appelle l'*ovales du cercle*, contenant le cercle. En nommant ces droites les *circulaires imaginaires* et leur point d'intersection, prenons pour les deux droites *circulaires imaginaires* comme quelconques. En nommant ces droites les *circulaires imaginaires* et leur point d'intersection, prenons pour les deux droites $x = 0, y = 0$, et pour les coordonnées z, w quelconques. P. Il est évident d'abord qu'une conique quelconque qui passe par les deux points P, Q doit être considérée comme un cercle, en nommant cette ligne quelconque transmutée à une conique quelconque ayant pour points de rebroussement les deux points P, Q , c'est-à-dire ayant les deux droites *circulaires imaginaires* pour tangentes au point de rebroussement, est la courbe du quatrième ordre ayant les deux droites *circulaires imaginaires* pour tangentes aux deux points de rebroussement.

M. Plücker: *Ueber solche Punkte die bei Curven höherer Ordnung als die zweiten den Brennpunkten der Kegelschnitte entsprechen*, *Journal de M. Crelle*, t. x. [1833] pp. 84–91). Cela étant, on cherche quelconque sera transmutée, ou simplement la transmutée de cette courbe, prenons les deux droites *circulaires imaginaires* pour points doubles celle entre les deux droites *circulaires imaginaires*. “Les ovals de Descartes ont deux points de rebroussement aux deux points Pülcker, et ces deux points doubles à l'infini de M. Plücker pour foyers communs.” Ainsi: “*Les ovals de Descartes* sont les courbes du quatrième ordre ayant pour foyers communs un point fixe, et à une conique qui a aussi ce point fixe pour un de ses foyers.” Donc, en résumé, tout cercle est transmuté en une courbe du quatrième ordre qui a les deux droites *circulaires imaginaires* pour points doubles, deux points doubles sont en effet les deux points de rebroussement à l'infini de M. Plücker, et la propriété des droites de se transmettre en limaçons (cas particulier), comme on sait, des ovals de Descartes. Donc, en résumé, tout cercle est transmuté en une courbe du quatrième ordre qui se réduit à la troisième classe au moyen des plans également avec le cas $a = \frac{1}{2}$. M. Roberts par ex. on trouve dans la Note citée de l'*Aperçu historique*, ans avoulument la propriété des droites de se transmettre en limaçons (cas particulier), comme on sait, des ovals de Descartes.